Lab Assignment

International Cybersecurity and Digital Forensics Academy

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

1. Purpose of the Lab Assignment

The purpose of this lab assignment is to investigate a Google Pixel 3 Android 10 phone for suspected unauthorised communication and information sharing. The investigation focuses on identifying device users and examining communication records. It also involves analysing messages, calls, photographs, and location-related evidence. The aim is to establish what happened using the available digital evidence. The findings will help determine whether the allegation is supported, contradicted, or unresolved.

2. Lab Report Structure

The report begins with an executive summary explaining the purpose and outcome of the investigation. It includes the investigation objectives, tools used, and evidence preservation procedures. The analysis and findings section covers device identity, communications, applications, photographs, and location records. A timeline is provided to show relevant events in chronological order.

2.1. Title Page

• Lab Title: Case Study 1: Google Pixel3 Android 10

• Student Name: Olanrewaju Wale Mustapha Oladimeji.

• Student ID: 2025/FWSD/11320.

• Course Name: SBT-DF205 Mobile and IoT Forensics Case Study

• Instructor Name: Aminu Idris

• Date of Submission: 20/10/2026

• Version:

2.2. Executive Summary

This lab assignment involves the forensic examination of a company issued Google Pixel 3 running Android 10. The investigation examines digital evidence relating to suspected unauthorised communication and information sharing. Tools such as SQLite and ExifTool are

used to analyse databases, messages, and media metadata. Relevant records are correlated to create a timeline of activities.

2.3. Lab Objectives

The objective is to preserve the original evidence and maintain its integrity throughout the investigation. It is also to identify device properties, user profiles, and relevant account associations. The investigation aims to examine SMS, MMS, calls, contacts, WhatsApp, and other application records. It also seeks to analyse photographs, location information, and timestamps to reconstruct relevant events. Finally, the objective is to develop a timeline and determine whether the allegation is supported, contradicted, or unresolved.

2.4. Tools and Resources Used

Kali Linux is used as the forensic examination environment for analysing the Android extraction.

SQLite and DB Browser for SQLite are used to examine application databases and communication records.

ExifTool is used to extract metadata from photographs and other media files.

Archive extraction utilities are used to access the supplied evidence while preserving the original archive. The Digital Corpora Android 10 Pixel 3 dataset and the supplied Android forensic investigation slides serve as the primary resources.

2.5. Methodology

Case brief

In this training case, Harbour Operations Ltd requests a forensic examination of a company-issued Google Pixel 3 following suspected unauthorised communication and sharing of information. As the examiner, establish the device users, reconstruct relevant communications and assess media and location evidence. Explain whether the records support, contradict or leave the allegation unresolved.

Use the original identities and historical dates in the supplied dataset. Base every finding on records you have examined.

Evidence and tools

Use Joshua Hickman's Android 10 file system extraction, hosted by Digital Corpora, in the Non-Cellebrite Extraction folder. The archive is approximately 5.25 GB and expands to approximately 9 GB.

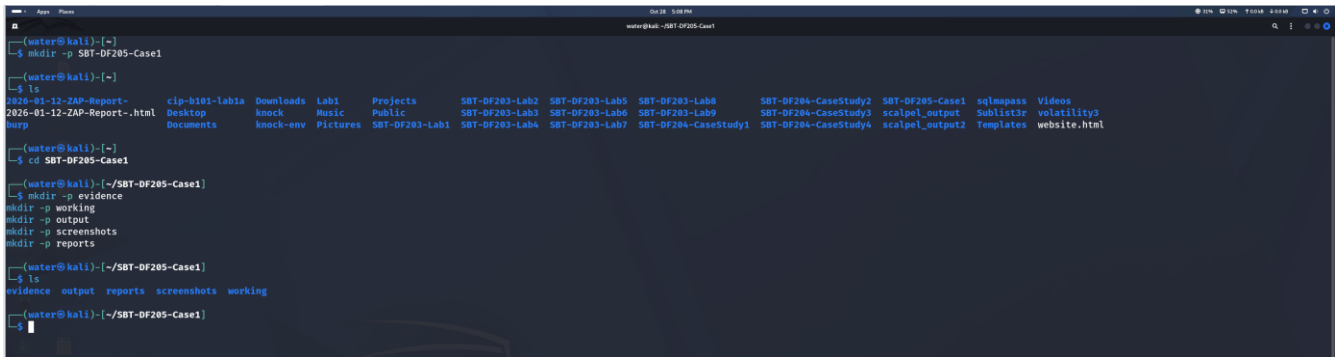
Download the Pixel 3 evidence archive

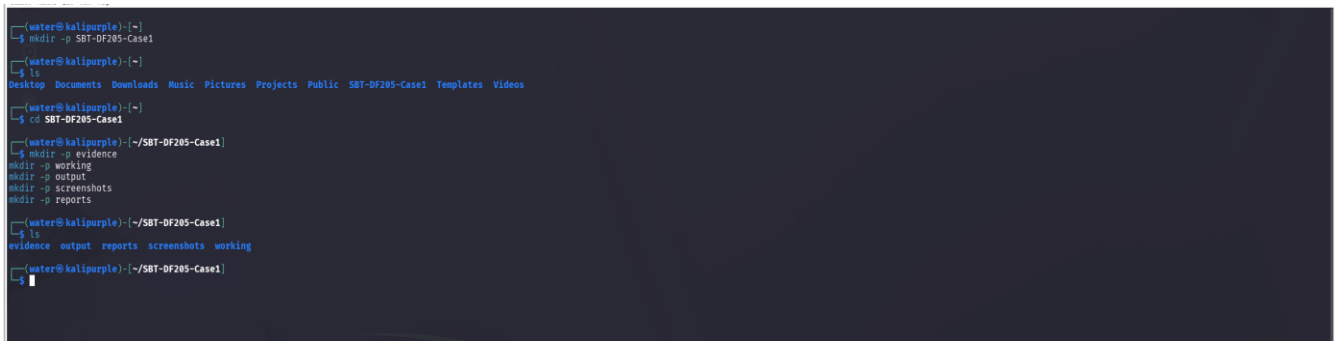
Required tools: an archive extractor, DB Browser for SQLite or SQLite, a text editor, ExifTool and a spreadsheet editor. Use Windows or Linux and allow at least 40 GB of free disk space. Refer to the supplied Android slides for the examination procedures.

Task 1 Evidence preservation

1. Create a handling log using case reference SBT-DF205-C1. Record the source URL, archive name, size, receipt time, examiner and tool versions.
2. Calculate the archive SHA-256 before extraction and recheck it at completion. Compare against a trusted manifest if supplied; otherwise record it as your baseline.
3. Maintain an evidence register for selected files, including evidence ID, source path, size and SHA-256. Record processing actions and preserve any database sidecars.

Creating the case folder





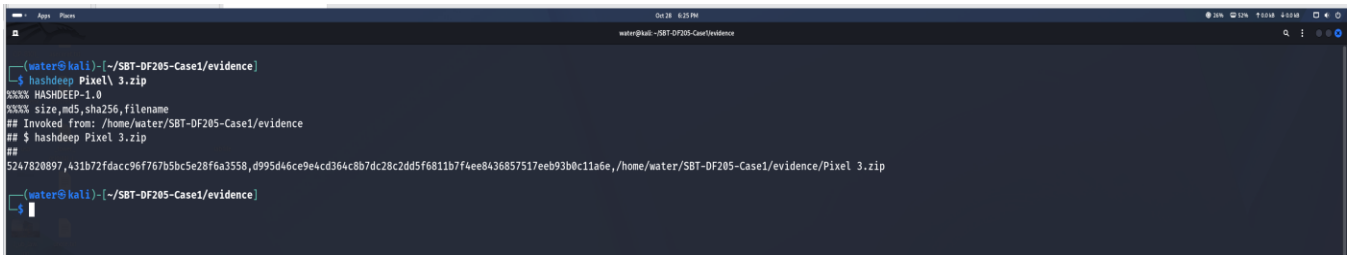
Dowloading the zip file







Verifying the hashes





%%%% HASHDEEP-1.0

%%%% size,md5,sha256,filename

Invoked from: /home/water/SBT-DF205-Case1/evidence

$ hashdeep Pixel 3.zip

##

5247820897,431b72fdacc96f767b5bc5e28f6a3558,d995d46ce9e4cd364c8b7dc28c2dd5f6811b7f4ee8436857517eeb93b0c11a6e,/home/water/SBT-DF205-Case1/evidence/Pixel 3.zip

%%%% HASHDEEP-1.0

%%%% size,md5,sha256,filename

Invoked from: /home/water/SBT-DF205-Case1/evidence
$ hashdeep Pixel 3.zip
##
5247820897,9cc37ebbbc4e918ee5427de1fe1deecc,ca6918ef8b20486b6a5ded15609ac51318f3
77829480f93be3ba15364a8aa00a,/home/water/SBT-DF205-Case1/evidence/Pixel 3.zip



sha256sum Pixel\ 3.zip
d995d46ce9e4cd364c8b7dc28c2dd5f6811b7f4ee8436857517eeb93b0c11a6e Pixel 3.zip
Unzip file into working copy









```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ l 'Pixel 3/data/system'
appops.xml              cachequota.xml           entropy.dat          locksettings.db     notification_policy.xml  package-dex-usage.list  package-watchdog.xml   shortcut_service.xml  uiderrors.txt
battery-history/        device_policies.xml      graphicsstats/       log-files.xml       overlays.xml             packages.list           procstats/             slice/                usagestats/
batterystats.bin        diskstats_cache.json     install_sessions.xml netpolicy.xml       package_cache/           packages-warnings.xml   recoverablekeystore.db sound_model.db        users/
batterystats-checkin.bin display-manager-state.xml job/               netstats/           package-cstats.list      packages.xml            screen_on_time         st_sound_model.db     watchlist_report.db
batterystats-daily.xml  dropbox/                 last-header.txt      notification_log.db package-dcl.list        package-usage.list      sensor_service/        sync/                 watchlist_settings.xml

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Task 2 Device identity and profiles

1. Use build.prop files to establish the manufacturer, model, Android version, build identifier and security patch level. Cite each property key.

2. Examine data/system/users to identify the user profiles. Identify at least two account or owner associations and corroborate one with another source.

3. Keep profile activity separate and explain the limits of attributing device activity to an individual.

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ cat 'Pixel 3/product/build.prop'

# begin common build properties
# autogenerated by build/make/tools/buildinfo_common.sh
ro.product.build.date=Tue Dec 17 19:34:12 UTC 2019
ro.product.build.date.utc=1576611252
ro.product.build.fingerprint=google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
ro.product.build.id=QQ1A.200105.003
ro.product.build.tags=release-keys
ro.product.build.type=user
ro.product.build.version.incremental=6079943
ro.product.build.version.release=10
ro.product.build.version.sdk=29
ro.product.product.brand=google
ro.product.product.device=blueline
ro.product.product.manufacturer=Google
ro.product.product.model=Pixel 3
ro.product.product.name=blueline
# end common build properties

#
# from device/google/crosshatch/product.prop
#

persist.rild.nitz_plmn=
persist.rild.nitz_long_ons_0=
persist.rild.nitz_long_ons_1=
persist.rild.nitz_long_ons_2=
persist.rild.nitz_long_ons_3=
persist.rild.nitz_short_ons_0=
persist.rild.nitz_short_ons_1=
persist.rild.nitz_short_ons_2=
persist.rild.nitz_short_ons_3=
DEVICE_PROVISIONED=1

# Set network mode to Global by default and no DSDS/DSDA
ro.telephony.default_network=10

debug.sf.hw=1
debug.gralloc.enable_fb_ubwc=1

# system props for the cne module
persist.vendor.cne.feature=1

# system props for the MM modules
media.stagefright.enable-player=true
media.stagefright.enable-http=true
media.stagefright.enable-aac=true
media.stagefright.enable-qcp=true
media.stagefright.enable-scan=true
mmp.enable.3g2=true
media.aac_51_output_enabled=true
mm.enable.smoothstreaming=true
#13631487 is decimal sum of supported codecs in AAL
#codecs:(PARSER_)AAC AC3 AMR_NB AMR_WB ASF AVI DTS FLV 3GP 3G2 MKV MP2PS MP2TS MP3 OGG QCP WAV FLAC AIFF APE DSD
mm.enable.qcom_parser=13631487
persist.mm.enable.prefetch=true

# Additional buffers shared between Camera and Video
vidc.enc.dcvs.extra-buff-count=2
vidc.enc.disable.pq=1

# system props for the data netmgrd
persist.data.netmgrd.qos.enable=true
persist.vendor.data.mode=concurrent

# system props for time-services
persist.timed.enable=true

# system props for perfetto
persist.traced.enable=1

# System prop to turn on CdmaLTEPhone always
telephony.lteOnCdmaDevice=1
```



```
#system prop for RmNet Data
persist.rmnet.data.enable=true
persist.data.wda.enable=true
persist.data.df.dl_mode=5
persist.data.df.ul_mode=5
persist.data.df.agg.dl_pkt=10
persist.data.df.agg.dl_size=4096
persist.data.df.mux_count=8
persist.data.df.iwlan_mux=9
persist.data.df.dev_name=rmnet_usb0

#
## system props for the data modules
#
#ro.use_data_netmgrd=true
persist.data.netmgrd.qos.enable=true
persist.data.mode=concurrent

# System props to enable Bluetooth Quality Report (BQR) feature
# BQR event mask configuration
persist.bluetooth.bqr.event_mask=14
# BQR minimum report interval configuration
persist.bluetooth.bqr.min_interval_ms=500

#
## Disable touch heatmap
#
ro.input.video_enabled=false
# end of device/google/crosshatch/product.prop
#
# ADDITIONAL PRODUCT PROPERTIES
#
ro.build.characteristics=nosdcard
persist.vendor.debug.sensors.accel_cal=1
masterclear.allow_retain_esim_profiles_after_fdr=true
ro.apk_verity.mode=1
ro.sys.sdcardfs=1
vendor.ese.loader_script_path=/sys/firmware/devicetree/base/soc/i2c@88c000/nq@28/ese/loader_scripts_path
ro.lmk.log_stats=true
ro.adb.secure=1
ro.lmk.low=1001
ro.lmk.medium=800
ro.lmk.critical=0
ro.lmk.critical_upgrade=false
ro.lmk.upgrade_pressure=100
ro.lmk.downgrade_pressure=100
ro.lmk.kill_heaviest_task=true
ro.lmk.kill_timeout_ms=100
ro.lmk.use_minfree_levels=true
ro.thermal_warmreset=true
ro.com.google.ime.height_ratio=1.2
ro.com.google.ime.bs_theme=true
ro.gfx.driver.0=com.google.pixel.crosshatch.gamedriver
ro.gfx.driver_build_time=1551427200
ro.opa.eligible_device=true
setupwizard.feature.baseline_setupwizard_enabled=true
ro.com.google.ime.theme_id=5
ro.com.android.dataroaming=false
ro.url.legal=http://www.google.com/intl/%s/mobile/android/basic/phone-legal.html
ro.url.legal.android_privacy=http://www.google.com/intl/%s/mobile/android/basic/privacy.html
ro.com.google.clientidbase=android-google
ro.error.receiver.system.apps=com.google.android.gms
ro.setupwizard.enterprise_mode=1
ro.atrace.core.services=com.google.android.gms,com.google.android.gms.ui,com.google.android.gms.persistent
ro.setupwizard.rotation_locked=true
setupwizard.enable_assist_gesture_training=true
ro.setupwizard.esim_cid_ignore=00000001
setupwizard.theme=glif_v3_light
setupwizard.feature.skip_button_use_mobile_data.carrier1839=true
setupwizard.feature.show_pai_screen_in_main_flow.carrier1839=false
setupwizard.feature.show_pixel_tos=true
ro.storage_manager.show_opt_in=false
ro.com.google.ime.system_lm_dir=/product/usr/share/ime/google/d3_lms
```

Checking the latest Security Patch



```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ grep patch 'Pixel 3/system/build.prop'
ro.build.version.security_patch=2020-01-01

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Checkin Mobile Carrier/Phone



```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ tree 'Pixel 3/data/user_de/0/com.android.providers.telephony/'
Pixel 3/data/user_de/0/com.android.providers.telephony/
├── databases
│   ├── carrierIdentification.db
│   ├── CarrierInformation.db
│   ├── cellbroadcasts.db
│   ├── HbpcdLookup.db
│   ├── mmssms.db
│   └── telephony.db
└── shared_prefs
    ├── build-id.xml
    ├── CarrierIdProvider.xml
    ├── preferred-apn.xml
    └── preferred-full-apn.xml

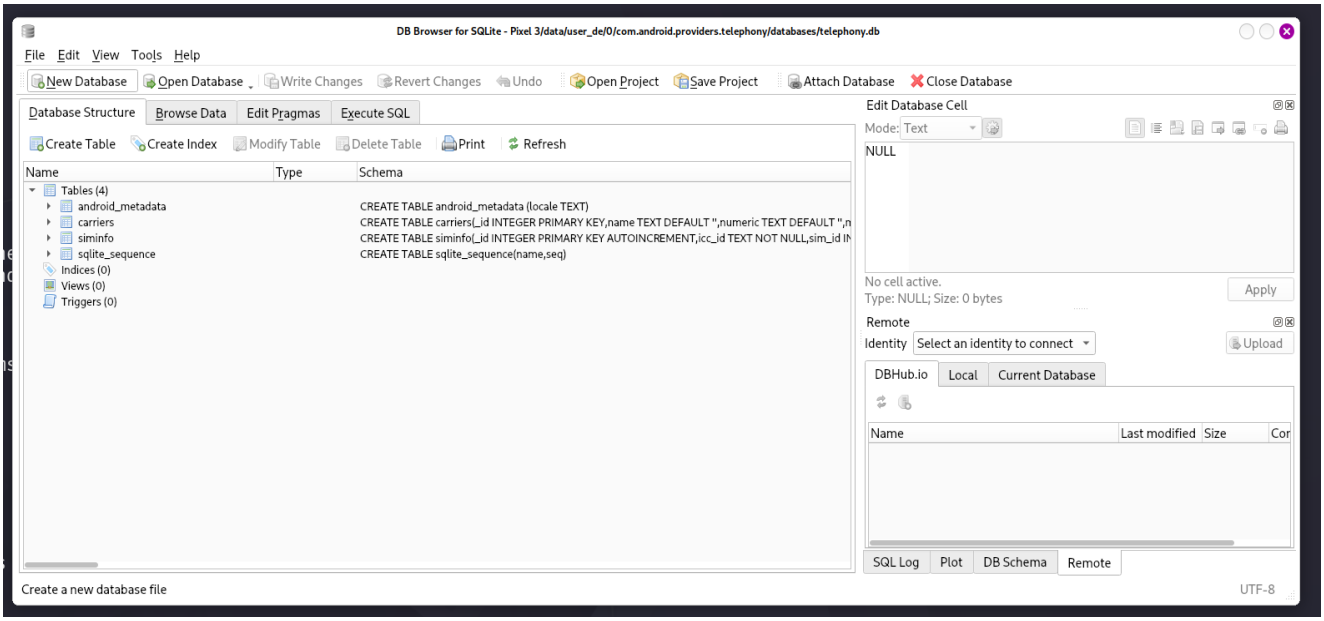
3 directories, 10 files

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Checking list of Telephony DB

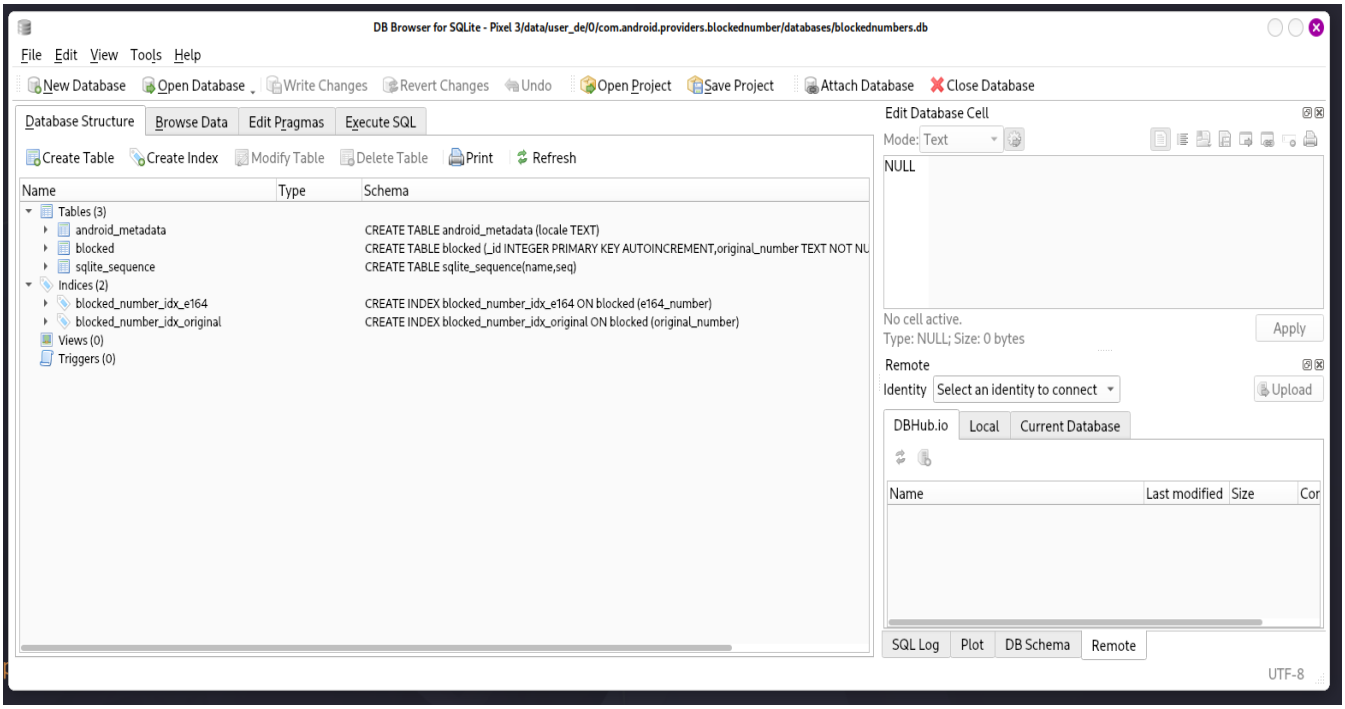


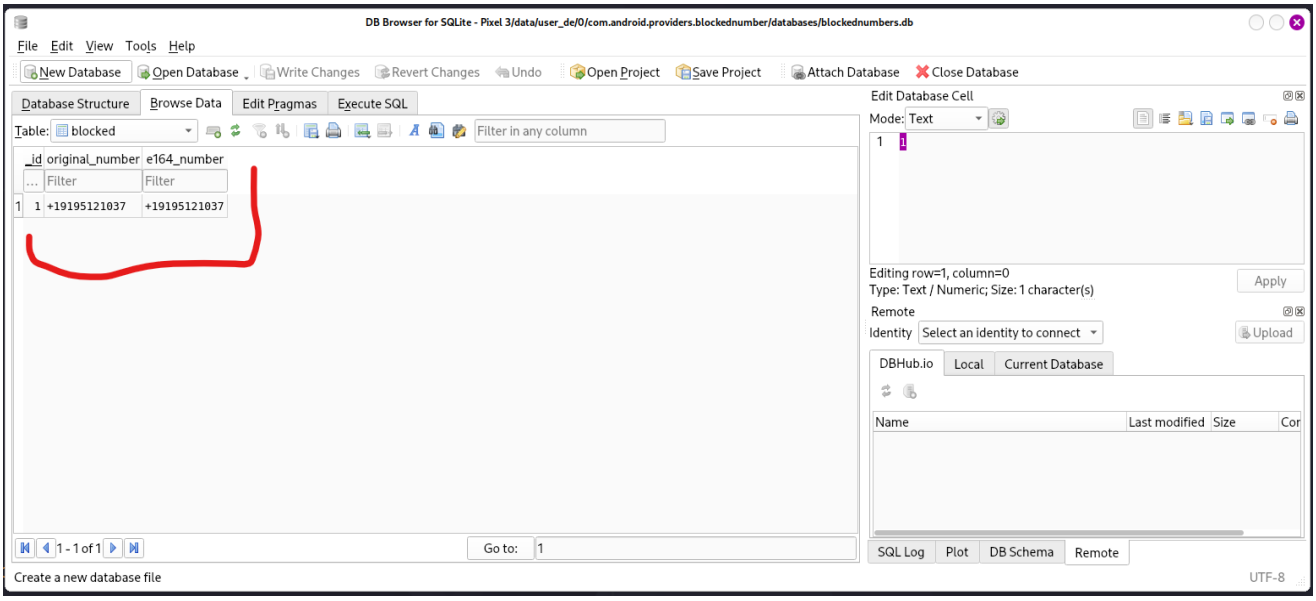
```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ sqlitebrowser 'Pixel 3/data/user_de/0/com.android.providers.telephony/databases/telephony.db'
```



Checking Blocked number

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ sqlitebrowser 'Pixel 3/data/user_de/0/com.android.providers.blockednumber/databases/blockednumbers.db'
```





Checking carrier Information

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ tree 'Pixel 3/data/user_de/0/com.android.phone/'
Pixel 3/data/user_de/0/com.android.phone/
├── files
│   ├── carrierconfig-com.google.android.apps.tycho-8901120300405680384 2-10015.xml
│   ├── carrierconfig-com.google.android.carrier-8901120300405680384 2-10015.xml
│   └── carrierconfig-com.google.android.carrier-8901260971148676693-10014.xml
└── shared_prefs
    ├── com.android.phone_preferences.xml
    └── _has_set_default_values.xml

3 directories, 5 files

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ cat 'Pixel 3/data/user_de/0/com.android.phone/files/carrierconfig-com.google.android.carrier-8901120300405680384 2-10015.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<carrier_config>
<package_version>8</package_version>
<bundle_data>
<string name="gps.lpp_profile">2</string>
<string-array name="enable_apps_string_array" num="0" />
<boolean name="carrier_app_required_during_setup_bool" value="true" />
<boolean name="config_telephony_use_own_number_for_voicemail_bool" value="true" />
<string name="carrier_config_version_string">fi_us-12000000333.16</string>
<string name="carrier_name_string">Google Fi</string>
<boolean name="carrier_name_override_bool" value="true" />
</bundle_data>
</carrier_config>

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ cat 'Pixel 3/data/user_de/0/com.android.phone/files/carrierconfig-com.google.android.carrier-8901260971148676693-10014.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<carrier_config>
<package_version>8</package_version>
<bundle_data>
<string name="gps.lpp_profile">2</string>
<string-array name="enable_apps_string_array" num="0" />
<boolean name="carrier_app_required_during_setup_bool" value="true" />
<boolean name="config_telephony_use_own_number_for_voicemail_bool" value="true" />
<string name="carrier_config_version_string">fi_us-12000000333.16</string>
<string name="carrier_name_string">Google Fi</string>
<boolean name="carrier_name_override_bool" value="true" />
</bundle_data>
</carrier_config>

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Checking Serial number of the device



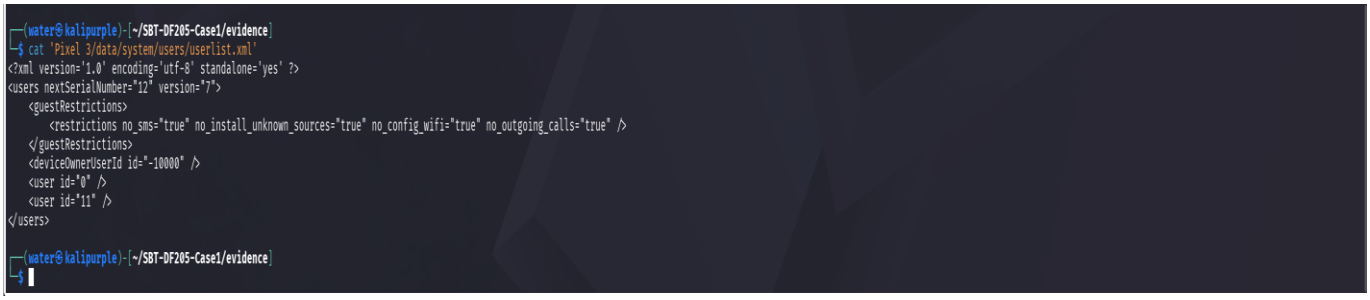
Checking serial number under vendor Folder



Checking User accounts on the device



Checking the owner of the device

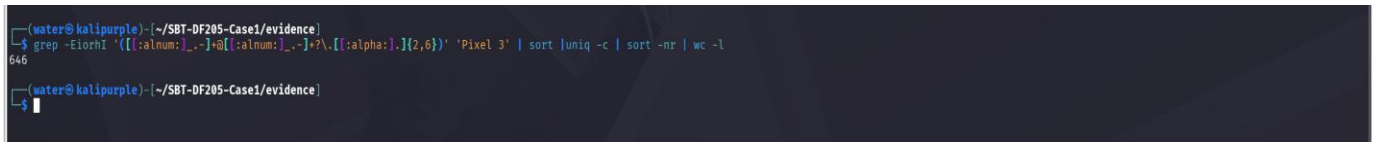


When did the device owner last time log in

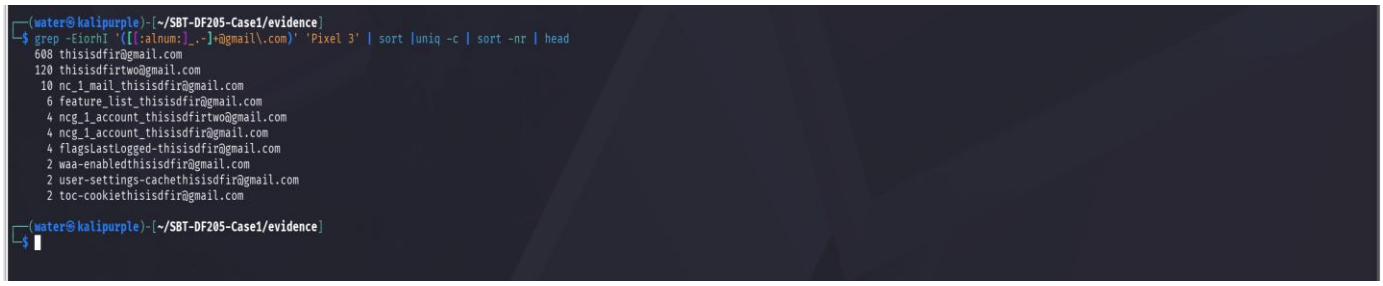


Email Address found on the image/Count of all emails





Checking all Gmail address appeared in the Phone



Gmail address registered for google service on Pixel 3



```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ tree 'Pixel 3/data/data/com.google.android.gm/shared_prefs' -L 1
Pixel 3/data/data/com.google.android.gm/shared_prefs
├── ACCOUNT_COUNT_MAP.xml
├── Account-thisisdfir@gmail.com.xml
├── AndroidMail.Main.xml
├── ANIMATABLE_LOGO_VIEW_FILE_KEY.xml
├── attachmentManager_-1294433372.xml
├── c9edm_-1294433372.xml
├── com.google.android.apps.gmail.accountDumpState&#58;thisisdfir@gmail.com&#58;com.google.xml
├── com.google.android.apps.gmail.dumpState.xml
├── com.google.android.apps.gmail.notifications&#58;thisisdfir@gmail.com.xml
├── com.google.android.gm_preferences.xml
├── com.google.android.gms.appid.xml
├── COMPOSE_UPLOADERS.xml
├── FirebaseAppHeartBeat.xml
├── FlagPrefs.xml
├── G6yPrefs.xml
├── GigSyncEngine_thisisdfir@gmail.com.xml
├── GMAIL_NETWORK_LOGGING_DEPOT.xml
├── Gmail.xml
├── _has_set_default_values.xml
├── MailAppProvider.xml
├── phenotype__com.google.android.libraries.social.populous.xml
├── timeZoneDbVersion.xml
├── triaged_notification_receive_times_thisisdfir@gmail.com.xml
├── UnifiedEmail.xml
├── warm_accounts_preferences.xml
├── watermark_notification_tokens_thisisdfir@gmail.com.xml
└── WebViewChromiumPrefs.xml

1 directory, 27 files

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ cat 'Pixel 3/data/data/com.google.android.gm/shared_prefs/Gmail.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <string name="thisisdfir-account-alias">thisisdfir</string>
    <string name="858233690-account-alias">858233690</string>
    <string name="TextNow-account-alias">TextNow</string>
    <string name="active-account">thisisdfir@gmail.com</string>
    <string name="removal-action">archive</string>
    <boolean name="conversation-overview-mode" value="true" />
    <string name="Signal-account-alias">Signal</string>
    <boolean name="conversation-list-swipe" value="true" />
    <boolean name="force_show_welcome_tour" value="false" />
    <boolean name="confirm-archive" value="false" />
    <boolean name="confirm-send" value="false" />
    <string name="thisisdfir@gmail.com-account-alias">thisisdfir@gmail.com</string>
    <string name="Skype-account-alias">Skype</string>
    <string name="+19195794674-account-alias">+19195794674</string>
    <string name="WhatsApp-account-alias">WhatsApp</string>
    <boolean name="confirm-delete" value="false" />
    <int name="welcome_tour_version" value="1" />
    <boolean name="mail-enable-threading" value="true" />
    <string name="cache-google-accounts-synced">thisisdfir@gmail.com</string>
</map>

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ cat 'Pixel 3/data/data/com.google.android.gm/shared_prefs/UnifiedEmail.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <int name="offline-search-index-corpus-version" value="2" />
    <boolean name="email-accounts-import-tried" value="true" />
    <long name="analytics-send-nb_accounts-epoch" value="1581644062878" />
    <int name="auto-advance-mode" value="0" />
    <int name="offline-search-index-schema-version" value="2" />
    <set name="display_images" />
    <boolean name="conversation-photo-teaser-shown-three" value="true" />
    <boolean name="sc-onboarding-shown-v2" value="true" />
    <boolean name="confirm-archive" value="false" />
    <string name="gm-density">DEFAULT</string>
    <long name="dots_animation_timestamp" value="1581643980962" />
    <long name="owners_name_cache_updated_timestamp" value="1581643980626" />
    <string name="email_address_prefix:thisisdfir@gmail.com">This Is DFIR</string>
    <string name="swipe-left-action">archive</string>
    <boolean name="dark-theme-active" value="true" />
    <int name="snap-header-mode" value="0" />
    <string name="default-reply-action">unset</string>
    <string name="conversation_source">show-conversation-from-previous-conversation</string>
    <boolean name="non-release-build" value="true" />
    <string name="custom-swipe-actions-onboarding-card-state">pending</string>
    <boolean name="conversation-overview-mode" value="true" />
    <set name="display_sender_images_patterns_set" />
    <boolean name="confirm-send" value="false" />
    <boolean name="conversation-list-sender-image" value="true" />
    <string name="swipe-right-action">archive</string>
    <boolean name="confirm-delete" value="false" />
    <int name="migrated-version" value="4" />
    <int name="prefs-version-number" value="4" />
</map>

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Gmail registered on the Playstore

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ pwd
/home/water/SBT-DF205-Case1/evidence

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ tree '/home/water/SBT-DF205-Case1/evidence/Pixel 3/data/data/com.android.vending/cache/logs/'
/home/water/SBT-DF205-Case1/evidence/Pixel 3/data/data/com.android.vending/cache/logs/
├── com.google.thisisdfir%40gmail.com.metalog
│   └── play_metalog.log
└── null_account.metalog
    ├── logs_upload_attempt.log
    └── play_metalog.log

3 directories, 3 files

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Task 3 Settings and connectivity

1. Examine available secure, global and system settings. Report two relevant settings with their stored keys and values.

2. Inspect Wi-Fi configuration and available Bluetooth information. Report supported identifiers or SSIDs and distinguish saved configuration from a proven connection at a specific time.

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ tree -L 1 'Pixel 3/data/system_ce/0/'
Pixel 3/data/system_ce/0/
├── accounts_ce.db
├── launch_params
├── recent_images
├── recent_tasks
├── shortcut_service
└── snapshots

6 directories, 1 file

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ tree -L 1 'Pixel 3/data/system_de/0/'
Pixel 3/data/system_de/0/
├── accounts_de.db
├── persisted_taskIds.txt
├── ringtones
└── spblob

3 directories, 2 files

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ sqlite3 'Pixel 3/data/system_de/0/accounts_de.db'
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> select name,type,datetime(last_password_entry_time_millis_epoch(1000, 'UNIXEPOCH') from accounts;
Parse error near line 1: near "from": syntax error
  ssword_entry_time_millis_epoch(1000, 'UNIXEPOCH') from accounts;
                                 error here ---^
sqlite> select name, type, datetime(last_password_entry_time_millis_epoch(1000, 'UNIXEPOCH') from accounts;
Parse error near line 2: near "from": syntax error
  sswrd_entry_time_millis_epoch(1000, 'UNIXEPOCH') from accounts;
                                 error here ---^
sqlite> select name, type, datetime(last_password_entry_time_millis_epoch/1000, 'UNIXEPOCH') from accounts;
```

name	type	datetime(last_pas ...
thisisdfir@gmail.com	com.google	2020-01-29 15:41:07
'858233690'	org.telegram.messenger	2020-01-29 18:31:04
Signal	org.thoughtcrime.securesms	2020-01-29 18:31:40
imo HD	com.imo.android.imous	2020-01-29 18:32:00
TikTok	com.zhiliaoapp.musically	2020-01-29 18:43:16
Messenger	com.facebook.messenger	2020-01-29 18:51:02
TDfir	com.twitter.android.auth.login	2020-01-29 19:42:51
WhatsApp	com.whatsapp	2020-01-29 21:10:02
thisisdfir	com.silentcircle.account	2020-02-01 20:41:31
Skype	com.skype.raider	2020-02-01 21:08:20
TextNow	com.enflick.android.TextNow.account	2020-02-08 19:54:52
'+19195794674'	com.viber.voip	2020-02-08 20:15:38
Duo	com.google.android.apps.tachyon	2020-02-14 01:19:15

```
sqlite>
```

Getting Bluetooth Name and Mac Address

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ grep -i 'bluetooth' 'Pixel 3/data/system/users/0/settings_secure.xml'
  <setting id="65" name="bluetooth_name" value="Pixel 3" package="android" defaultValue="Pixel 3" defaultSysSet="true" />
  <setting id="71" name="bluetooth_address" value="7C:D9:5C:AC:A2:CE" package="android" defaultValue="7C:D9:5C:AC:A2:CE" defaultSysSet="true" />

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Checking the Wifi service set identifier (SSID)

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ cat 'Pixel 3/data/misc/wifi/WifiConfigStore.xml'
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<WifiConfigStoreData>
<int name="Version" value="2" />
<Integrity>
<byte-array name="EncryptedData" num="48">e71394aa95f2d767a1c46e787ca99c0478e12467241553a8551005969fd38294e8a4082529d69a15d5bc49f28ca21a54</byte-array>
<byte-array name="IV" num="12">18234dad790da5cb3ba8eceb</byte-array>
</Integrity>
<NetworkList />
<MacAddressMap>
<map name="MacMapEntry">
<string name="&quot;CcookiesDcastleR5&quot;WPA_PSK">ae:f6:c7:a8:a5:c2</string>
</map>
</MacAddressMap>
<PasspointConfigData>
<long name="ProviderIndex" value="0" />
</PasspointConfigData>
</WifiConfigStoreData>

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Task 4 AOSP communications and contacts

1. Compare Telephony Provider copies of mmssms.db by path, hash and row counts. Start under data/data/com.android.providers.telephony/databases; inspect user_de copies separately because their contents may differ.
2. Extract at least five SMS records and reconstruct one conversation. Include record IDs, addresses, content, direction, read flags, raw times and validated UTC conversions. Reconstruct one MMS using pdu and part, including text or attachment references.
3. Use contacts2.db and calllog.db to link at least two communication numbers to contacts where matches exist, and examine at least three calls.
4. Inspect the Calendar Provider database and report relevant table names and row counts. Document any empty tables.



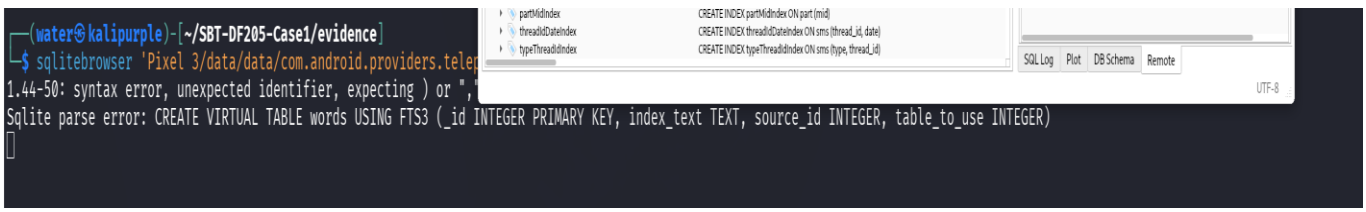
Checking the message service folder structure

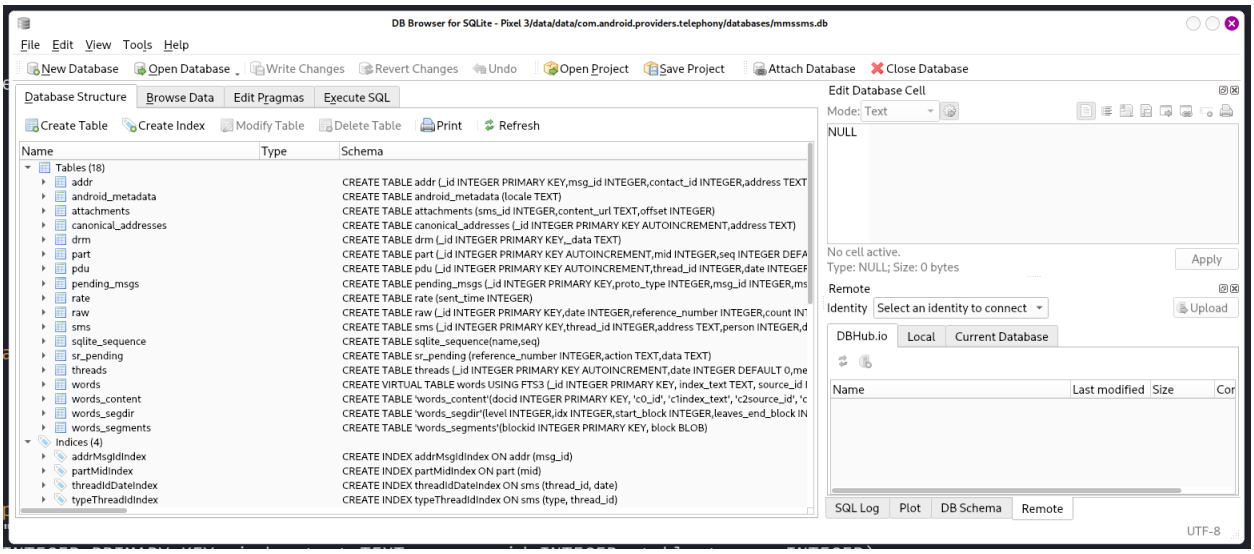


Checking Location of mmssms.db

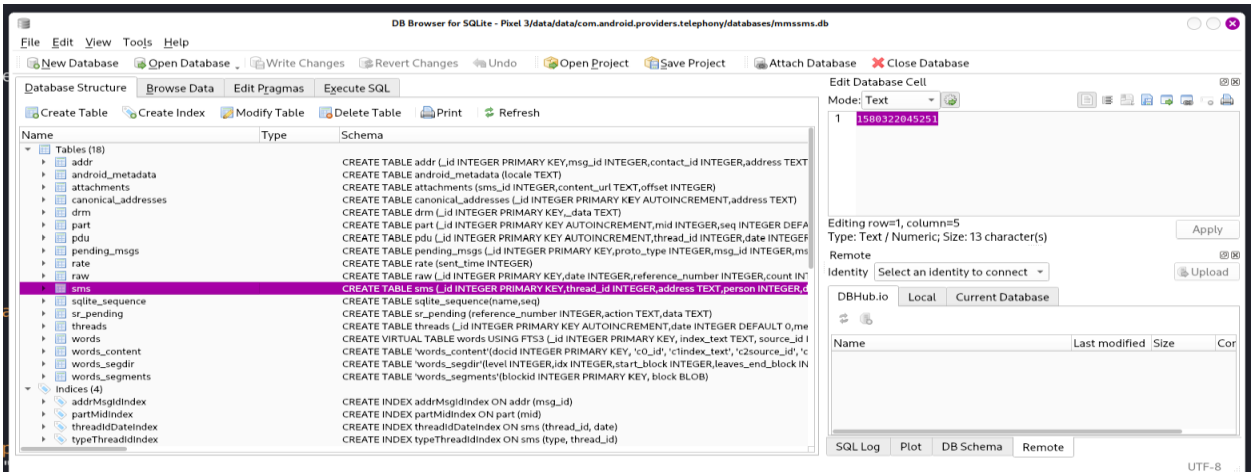


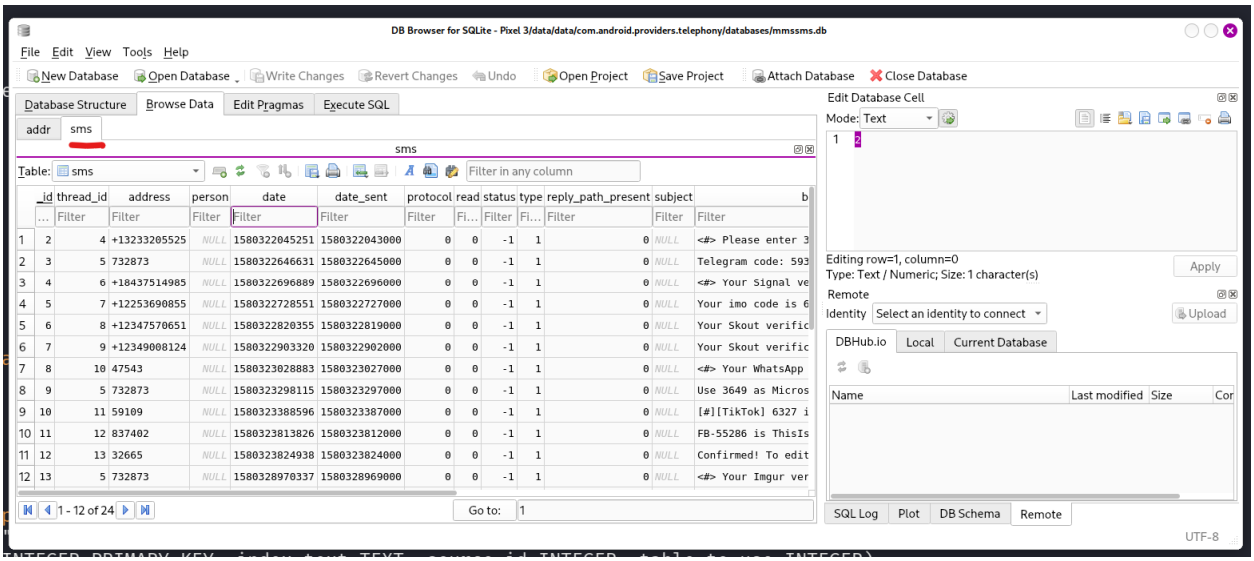
Summary of tables in mmssms.db





Investigating the sms table

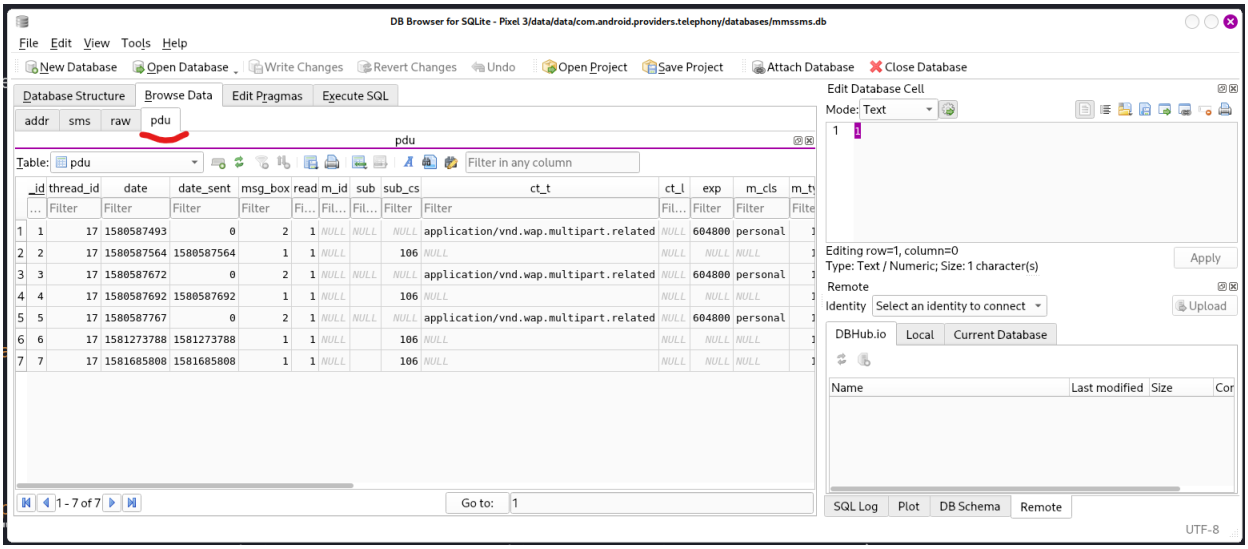




```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$ date -d '@1580322045251' -u
Sat May 30 09:54:11 AM UTC 52048

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence]
└─$
```

Investigating pdu



Contact Management Service

```
┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence/Pixel 3/data]
└─$ ls
adb  app      cache         data  magisk_backup_e132ca5af8732354d2f63dcfcd4331cddc309ae8  misc     nfc       resource-cache  system_ce  tombstones   user_de  vendor_ce
apex  backup  dalvik-cache  drm   media                                                  misc_ce  property  system          system_de  unencrypted  vendor

┌──(water㉿kalipurple)-[~/SBT-DF205-Case1/evidence/Pixel 3/data]
└─$ cd data

┌──(water㉿kalipurple)-[~/../evidence/Pixel 3/data/data]
└─$ ls
com.android.chrome                          com.facebook.orca                          com.google.android.apps.turbo          com.google.android.GoogleCamera          com.google.android.webview          com.skype.raider
com.android.hotwordenrollment.okgoogle      com.google.android.apps.chromecast.app     com.google.android.apps.tycho          com.google.android.googlequicksearchbox  com.google.android.wfcactivation    com.snapchat.android
com.android.hotwordenrollment.xgoogle       com.google.android.apps.docs               com.google.android.apps.walletnfcrel   com.google.android.gsf                   com.google.android.youtube          com.spotify.music
com.android.keychain                        com.google.android.apps.docs.editors.docs  com.google.android.apps.wallpaper      com.google.android.ims                   com.google.ar.core                  com.thinkyeah.galleryvault
com.android.mtp                             com.google.android.apps.dreamliner         com.google.android.apps.wellbeing      com.google.android.inputmethod.latin     com.google.intelligence.sense       com.topjohnwu.magisk
com.android.nfc                             com.google.android.apps.enterprise.dmagent com.google.android.apps.work.oobconfig com.google.android.keep                    com.google.vr.apps.ornament         com.twitter.android
com.android.providers.calendar              com.google.android.apps.gcs                com.google.android.apps.youtube.music  com.google.android.markup                com.google.vr.vrcore                com.venmo
com.android.providers.contacts              com.google.android.apps.helprtc            com.google.android.as                  com.google.android.music                 com.imgur.mobile                    com.viber.voip
com.android.providers.downloads             com.google.android.apps.magazines          com.google.android.calendar            com.google.android.onetimeinitializer    com.imo.android.imous               com.whatsapp
com.android.providers.media                 com.google.android.apps.maps               com.google.android.configupdater       com.google.android.partnersetup          com.instagram.android               com.wire
com.android.providers.telephony             com.google.android.apps.messaging          com.google.android.contacts            com.google.android.pixel.setupwizard     com.instagram.threadsapp            com.zhiliaoapp.musically
com.android.providers.userdictionary        com.google.android.apps.nexuslauncher      com.google.android.deskclock           com.google.android.projection.gearhead   com.mewe                            jp.naver.line.android
com.android.service.ims.presence            com.google.android.apps.photos             com.google.android.dialer              com.google.android.settings.intelligence com.mywickr.wickr2                  kik.android
com.android.traceur                         com.google.android.apps.pixelmigrate       com.google.android.documentsui         com.google.android.setupwizard           com.qti.ltebc                       org.telegram.messenger
com.android.vending                         com.google.android.apps.safetyhub          com.google.android.ext.services        com.google.android.syncadapters.contacts com.radicalapps.cyberdust           org.thoughtcrime.securesms
com.discord                                 com.google.android.apps.tachyon            com.google.android.gm                  com.google.android.tts                   com.silentcircle.silentphone        org.torproject.torbrowser
com.enflick.android.TextNow                 com.google.android.apps.tips               com.google.android.gms                 com.google.android.videos                com.skout.android

┌──(water㉿kalipurple)-[~/../evidence/Pixel 3/data/data]
└─$
```

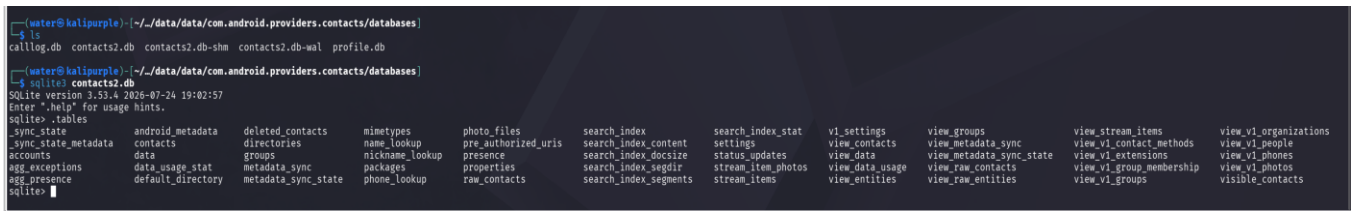
```
┌──(water㉿kalipurple)-[~/../evidence/Pixel 3/data/data]
└─$ tree -L 2 com.android.providers.contacts
com.android.providers.contacts
├── app_voicemail-data
│   ├── voicemail4257613984156813452
│   └── voicemail6279788559607275412
├── databases
│   ├── calllog.db
│   ├── contacts2.db
│   └── profile.db
└── shared_prefs
    ├── com.android.providers.contacts_preferences.xml
    └── ContactsUpgradeReceiver.xml

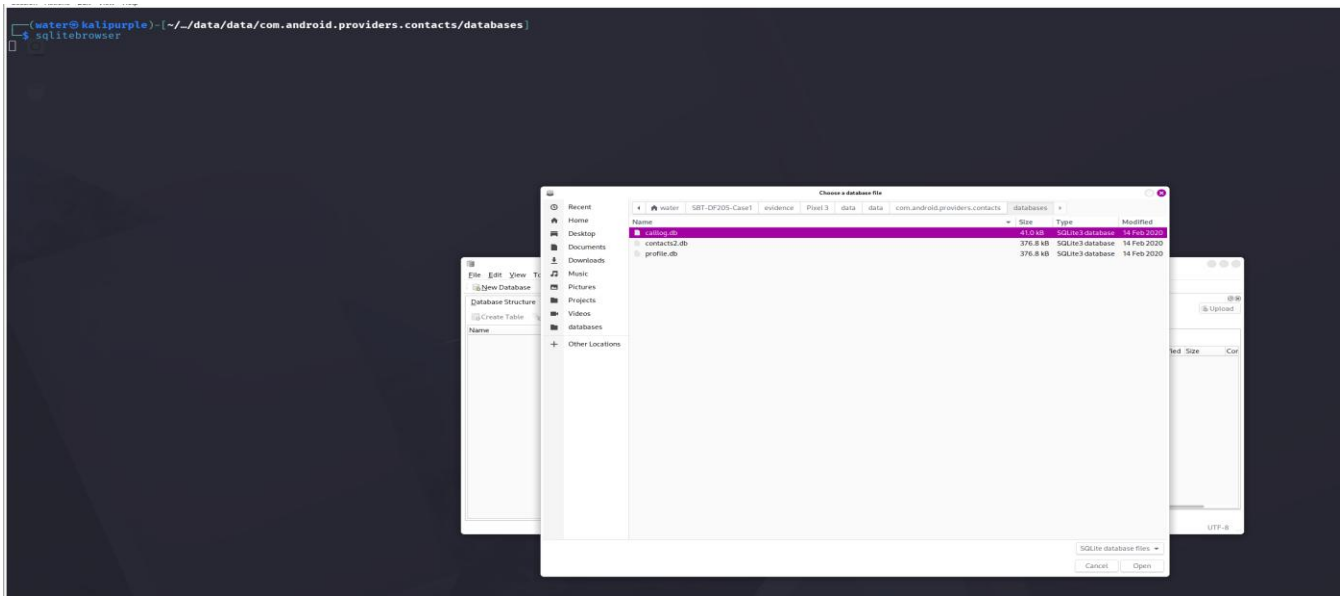
4 directories, 7 files

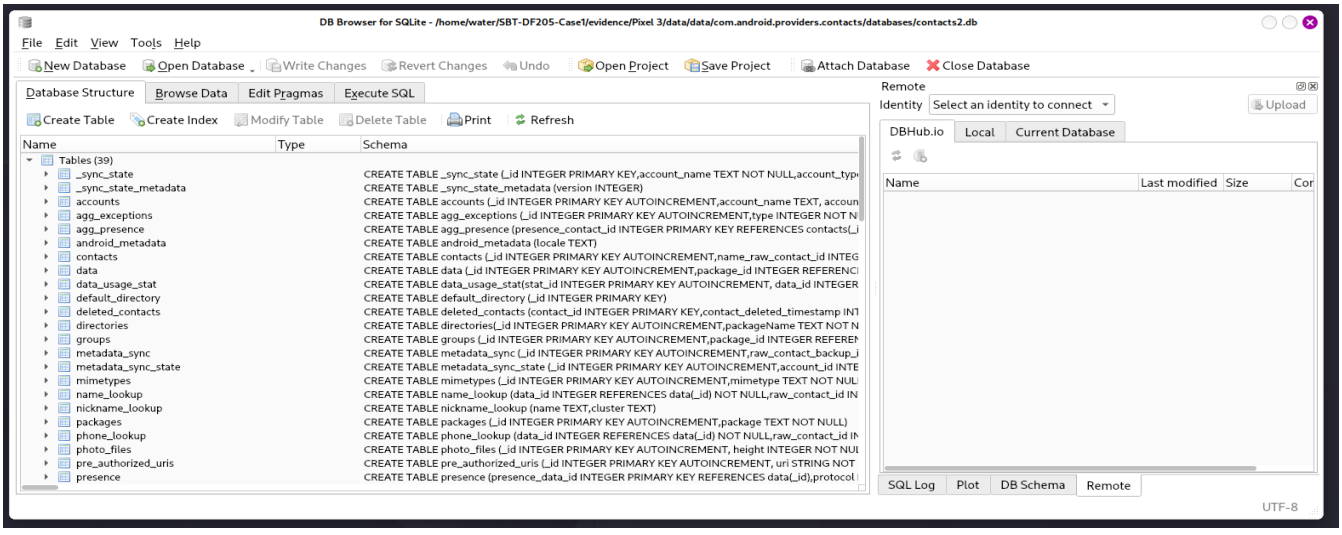
┌──(water㉿kalipurple)-[~/../evidence/Pixel 3/data/data]
└─$
```

Checking tables in contacts.db

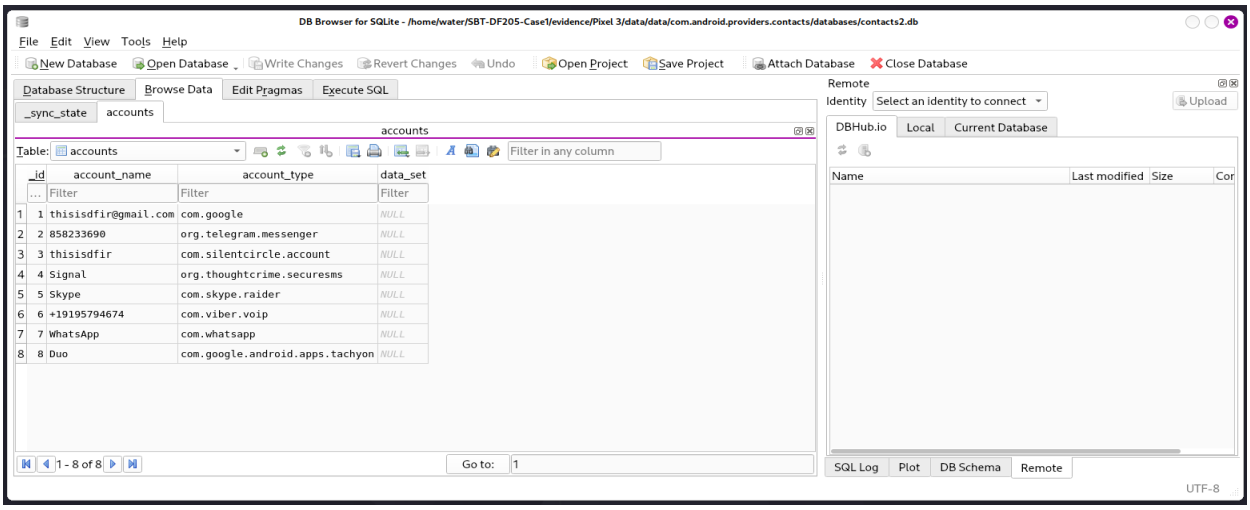




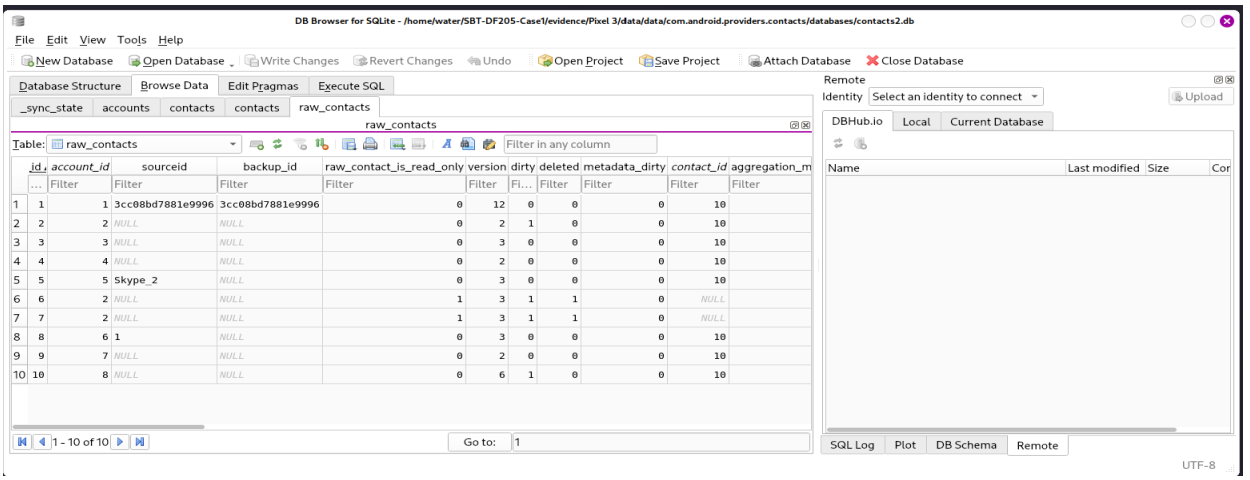


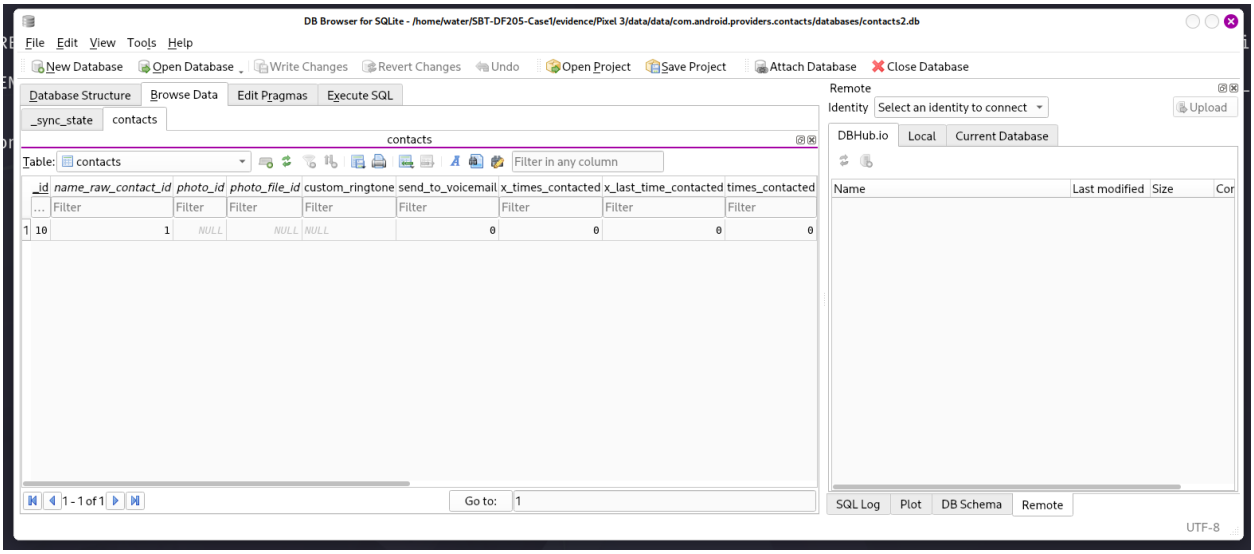


Checking Account tables



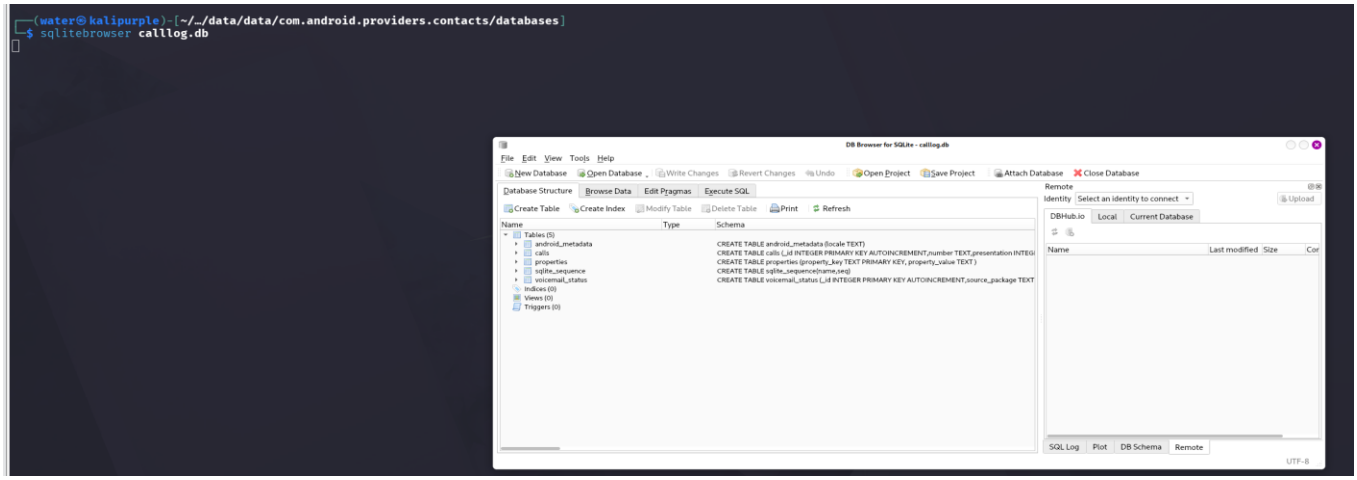
Raw contact table

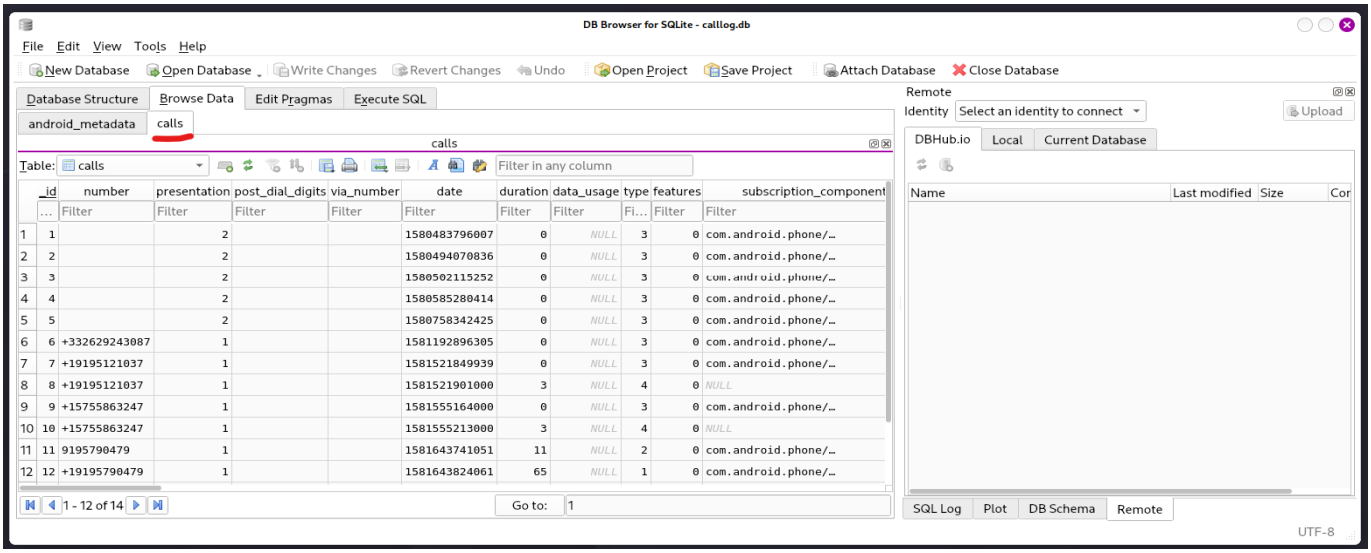




Call log.db

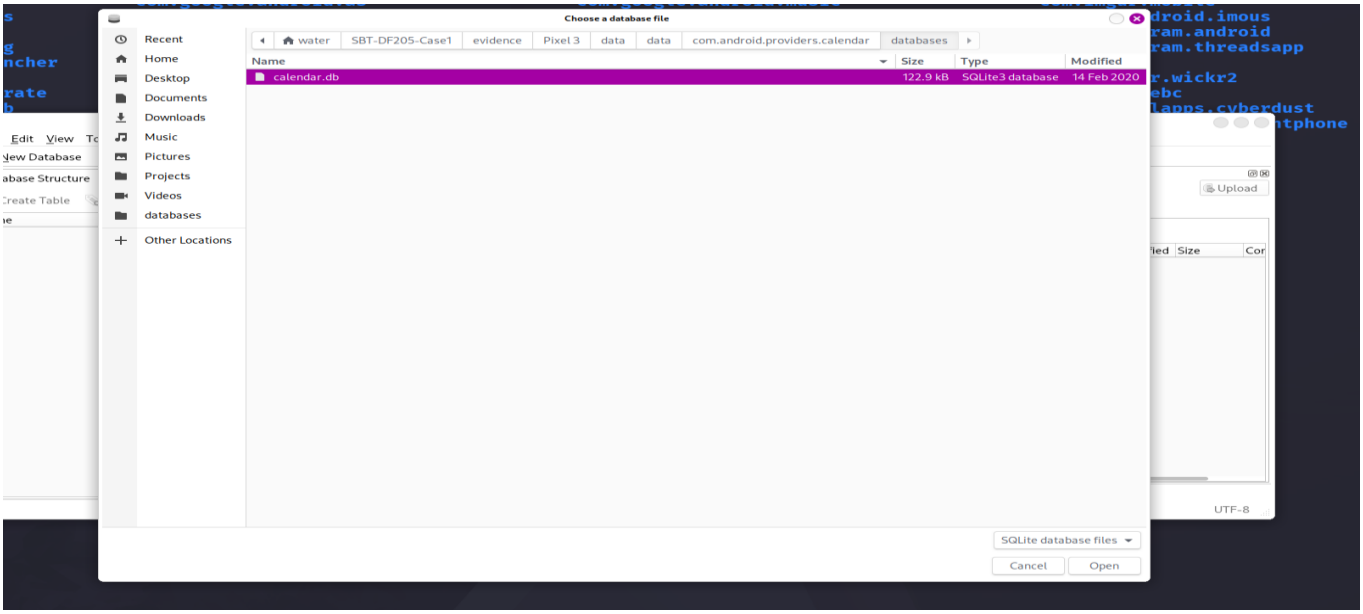

(water㉿kalipurple)-[~/…/data/data/com.android.providers.contacts/databases]
$ tree
.
├── calllog.db
├── contacts2.db
└── profile.db
1 directory, 3 files
(water㉿kalipurple)-[~/…/data/data/com.android.providers.contacts/databases]
$

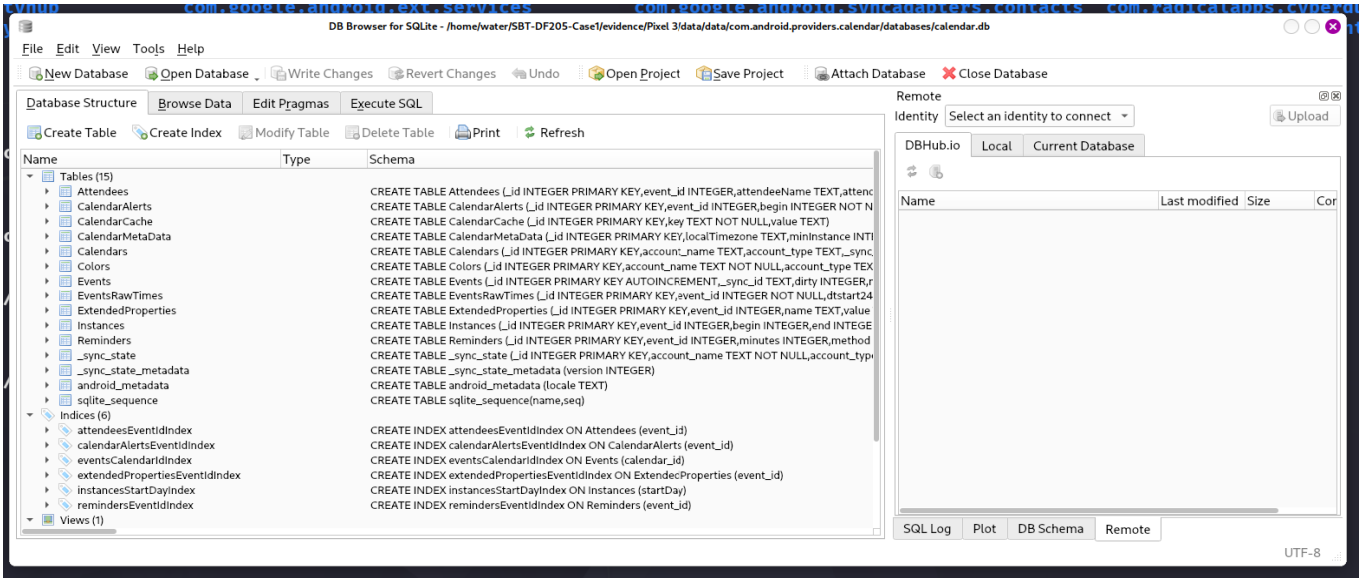

(water㉿kalipurple)-[~/…/data/data/com.android.providers.contacts/databases]
$ sqlitebrowser calllog.db
DB Browser for SQLite - calllog.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Delete Table
Print
Refresh
Name
Type
Schema
Tables (5)
android_metadata
calls
properties
sqlite_sequence
voicemail_status
Indices (0)
Views (0)
Triggers (0)
CREATE TABLE android_metadata (locale TEXT)
CREATE TABLE calls (_id INTEGER PRIMARY KEY AUTOINCREMENT,number TEXT,presentation INTEGI
CREATE TABLE properties (property_key TEXT PRIMARY KEY, property_value TEXT)
CREATE TABLE sqlite_sequence(name,seq)
CREATE TABLE voicemail_status (_id INTEGER PRIMARY KEY AUTOINCREMENT,source_package TEXT
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
Cor
SQL Log
Plot
DB Schema
Remote
UTF-8


DB Browser for SQLite - calllog.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
android_metadata
calls
calls
Table: calls
Filter in any column
_id number presentation post_dial_digits via_number date duration data_usage type features subscription_component
1 1 2 1580483796007 0 NULL 3 0 com.android.phone/…
2 2 2 1580494070836 0 NULL 3 0 com.android.phone/…
3 3 2 1580502115252 0 NULL 3 0 com.android.phone/…
4 4 2 1580585280414 0 NULL 3 0 com.android.phone/…
5 5 2 1580758342425 0 NULL 3 0 com.android.phone/…
6 6 +332629243087 1 1581192896305 0 NULL 3 0 com.android.phone/…
7 7 +19195121037 1 1581521849939 0 NULL 3 0 com.android.phone/…
8 8 +19195121037 1 1581521901000 3 NULL 4 0 NULL
9 9 +15755863247 1 1581555164000 0 NULL 3 0 com.android.phone/…
10 10 +15755863247 1 1581555213000 3 NULL 4 0 NULL
11 11 9195790479 1 1581643741051 11 NULL 2 0 com.android.phone/…
12 12 +19195790479 1 1581643824061 65 NULL 1 0 com.android.phone/…
1 - 12 of 14
Go to: 1
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
Cor
SQL Log
Plot
DB Schema
Remote
UTF-8

Calender Folder Structure



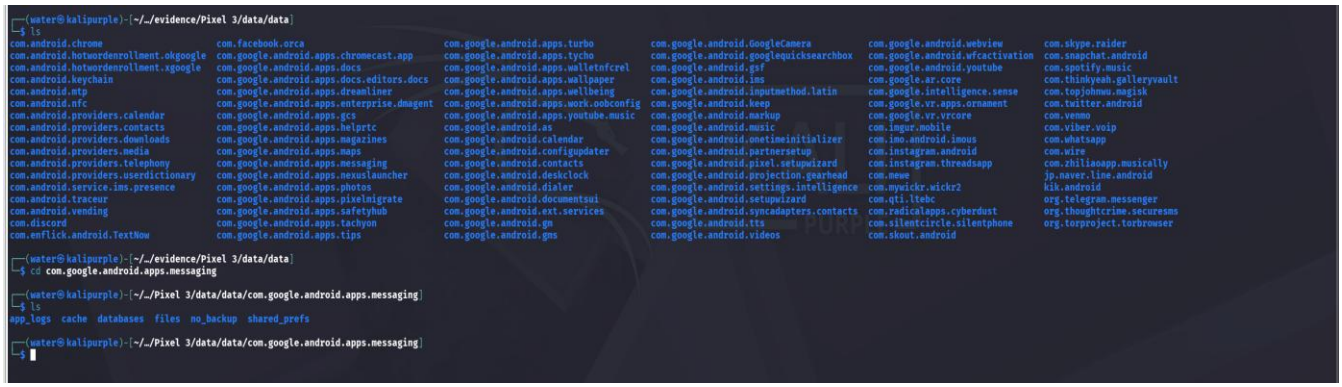


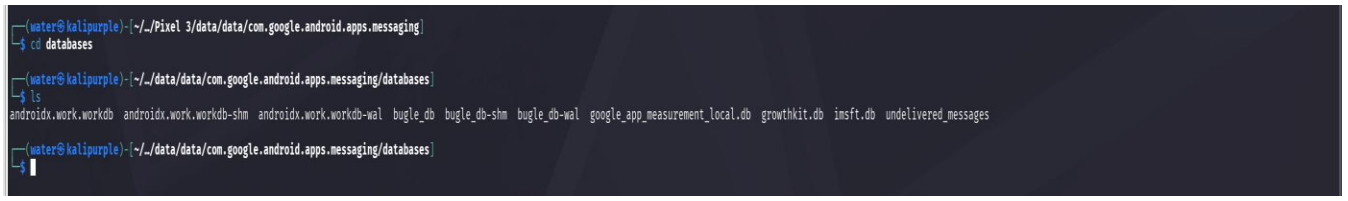


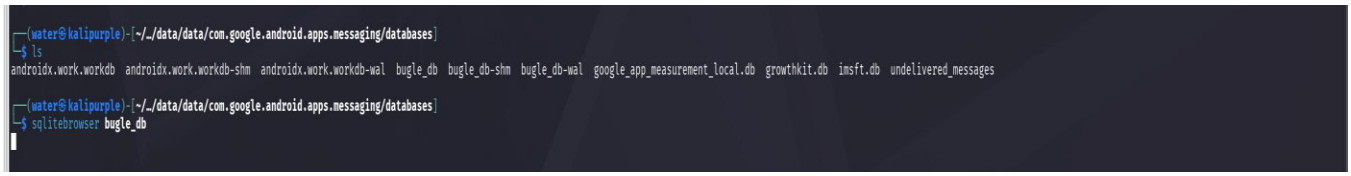
Task 5 Google Messages and Dialer

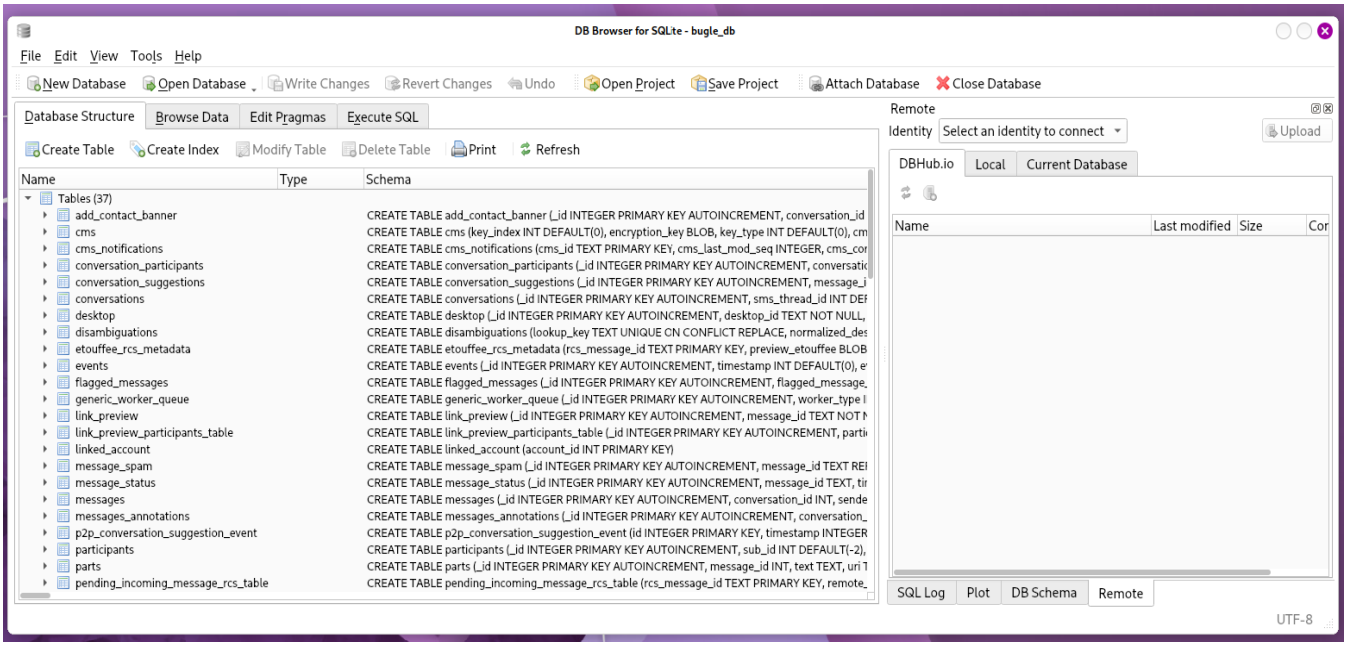
1. Examine Google Messages bugle_db with its sidecars. Match at least three SMS records to the Telephony Provider using record links or matching content, address and time.

2. Compare at least two Google Dialer calls with the AOSP call log where matches exist. Report differences and reproduce one important result using SQL or direct source examination.

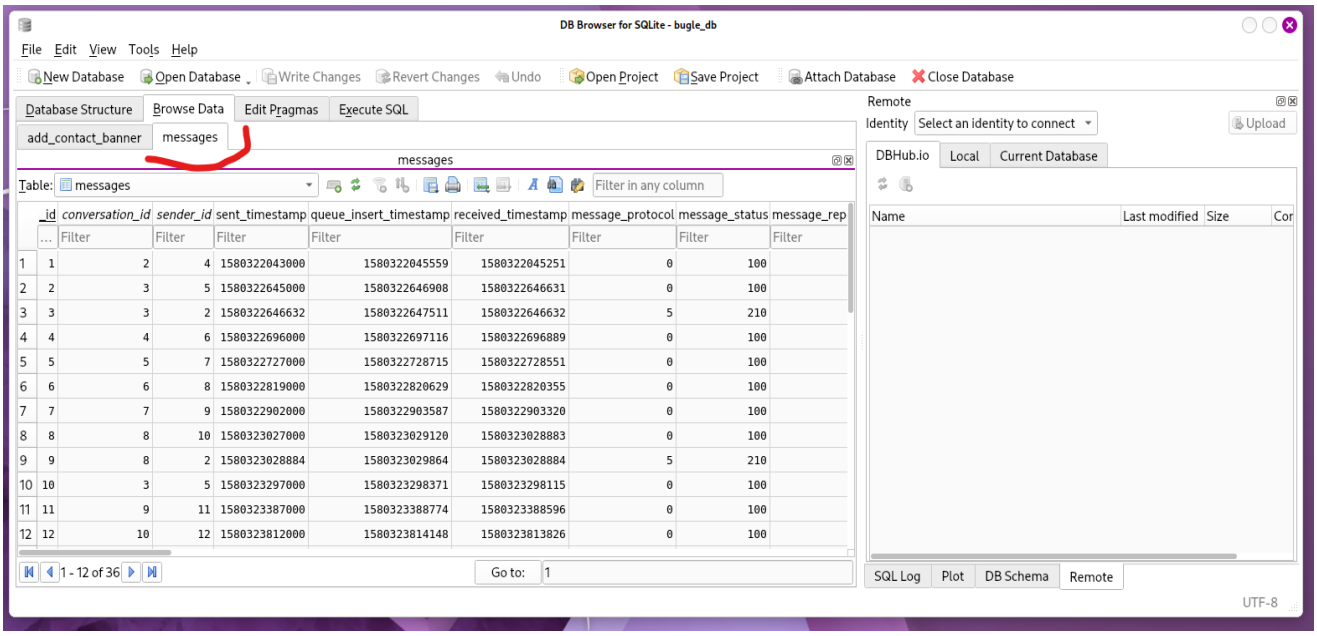




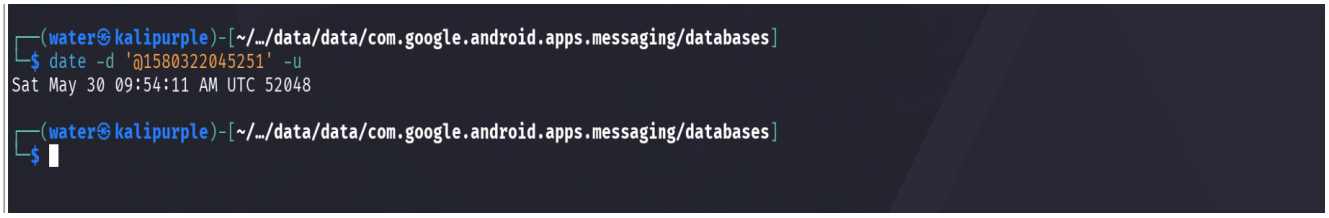




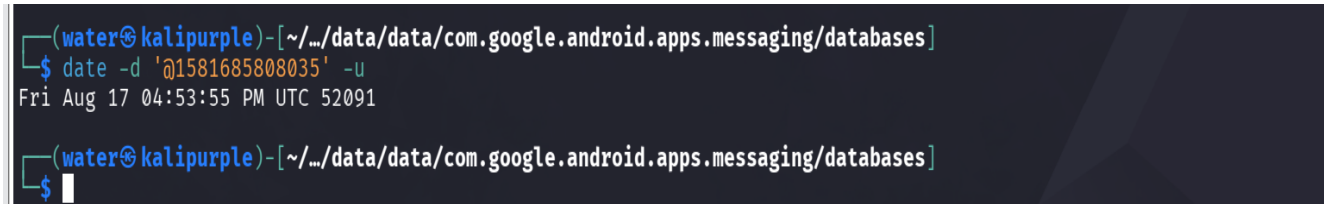
Messages Table



Received Timestamp



Sent Timestamp



Task 6 WhatsApp and another application

1. Use WhatsApp preferences, msgstore.db and wa.db to identify the account and reconstruct an exchange containing at least two messages. Record participants, direction, content or media references and timestamps.

2. Choose TextNow or Kik. Identify an account association and extract at least two communication records where available. Correlate one participant with another source.

3. Inspect the actual schema and validate message status meanings for the application version in the evidence.

```
┌──(water㉿kalipurple)-[~/…/evidence/Pixel 3/data/data]
└─$ ls
com.android.chrome                          com.facebook.orca                           com.google.android.apps.turbo               com.google.android.GoogleCamera             com.google.android.webview                  com.skype.raider
com.android.hotwordenrollment.okgoogle      com.google.android.apps.chromecast.app      com.google.android.apps.tycho               com.google.android.googlequicksearchbox     com.google.android.wfcactivation            com.snapchat.android
com.android.hotwordenrollment.xgoogle       com.google.android.apps.docs                com.google.android.apps.walletnfcrel        com.google.android.gsf                      com.google.android.youtube                  com.spotify.music
com.android.keychain                        com.google.android.apps.docs.editors.docs   com.google.android.apps.wallpaper           com.google.android.ims                      com.google.ar.core                          com.thinkyeah.galleryvault
com.android.mtp                             com.google.android.apps.dreamliner          com.google.android.apps.wellbeing           com.google.android.inputmethod.latin        com.google.intelligence.sense               com.topjohnwu.magisk
com.android.nfc                             com.google.android.apps.enterprise.dmagent  com.google.android.apps.work.oobconfig      com.google.android.keep                     com.google.vr.apps.ornament                 com.twitter.android
com.android.providers.calendar              com.google.android.apps.gcs                 com.google.android.apps.youtube.music       com.google.android.markup                   com.google.vr.vrcore                        com.venmo
com.android.providers.contacts              com.google.android.apps.helprtc             com.google.android.as                       com.google.android.music                    com.imgur.mobile                            com.viber.voip
com.android.providers.downloads             com.google.android.apps.magazines           com.google.android.calendar                 com.google.android.onetimeinitializer       com.imo.android.imous                       com.whatsapp
com.android.providers.media                 com.google.android.apps.maps                com.google.android.configupdater            com.google.android.partnersetup             com.instagram.android                       com.wire
com.android.providers.telephony             com.google.android.apps.messaging           com.google.android.contacts                 com.google.android.pixel.setupwizard        com.instagram.threadsapp                    com.zhiliaoapp.musically
com.android.providers.userdictionary        com.google.android.apps.nexuslauncher       com.google.android.deskclock                com.google.android.projection.gearhead      com.mewe                                    jp.naver.line.android
com.android.service.ims.presence            com.google.android.apps.photos              com.google.android.dialer                   com.google.android.settings.intelligence    com.mywickr.wickr2                          kik.android
com.android.traceur                         com.google.android.apps.pixelmigrate        com.google.android.documentsui              com.google.android.setupwizard              com.qti.ltebc                               org.telegram.messenger
com.android.vending                         com.google.android.apps.safetyhub           com.google.android.ext.services             com.google.android.syncadapters.contacts    com.radicalapps.cyberdust                   org.thoughtcrime.securesms
com.discord                                 com.google.android.apps.tachyon             com.google.android.gm                       com.google.android.tts                      com.silentcircle.silentphone                org.torproject.torbrowser
com.enflick.android.TextNow                 com.google.android.apps.tips                com.google.android.gms                      com.google.android.videos                   com.skout.android

┌──(water㉿kalipurple)-[~/…/evidence/Pixel 3/data/data]
└─$ cd com.whatsapp

┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/com.whatsapp]
└─$ ls
cache  databases  files  no_backup  shared_prefs

┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/com.whatsapp]
└─$ ll
total 20
drwxrwxr-x 4 water water 4096 Feb 15  2020 cache
drwxrwxr-x 2 water water 4096 Feb 15  2020 databases
drwxrwxr-x 6 water water 4096 Feb 15  2020 files
drwxrwxr-x 2 water water 4096 Feb 15  2020 no_backup
drwxrwxr-x 2 water water 4096 Feb 15  2020 shared_prefs

┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/com.whatsapp]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/com.whatsapp]
└─$ cd shared_prefs


┌──(water㉿kalipurple)-[~/…/data/data/com.whatsapp/shared_prefs]
└─$ tree -L 1 .
.
├── ab-props.xml
├── chatCounts.xml
├── com.google.android.gms.appid.xml
├── com.google.maps.api.android.lib6.drd.PREFERENCES_FILE.xml
├── com.whatsapp_payment_preferences.xml
├── com.whatsapp_preferences_light.xml
├── com.whatsapp_preferences.xml
├── contact_sync_prefs.xml
├── field-stats-events-sampling.xml
├── _has_set_default_values.xml
├── keystore.xml
├── MapviewInitializerPreferences.xml
├── media_bandwidth_shared_preferences_v2.xml
├── media_daily_usage_preferences_v1.xml
├── registration.RegisterPhone.xml
├── registration.VerifySms.xml
├── time_spent_prefs.xml
└── voip_prefs.xml

1 directory, 18 files

┌──(water㉿kalipurple)-[~/…/data/data/com.whatsapp/shared_prefs]
└─$
```

Database (msgstore.db)

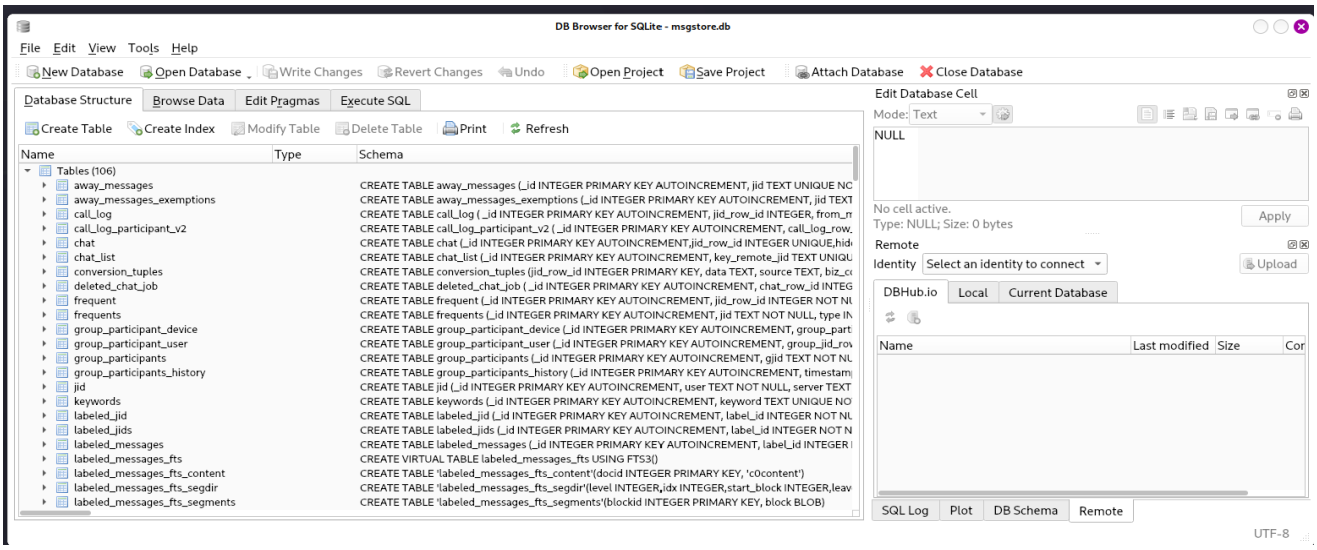
```
┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/com.whatsapp]
└─$ cd databases

┌──(water㉿kalipurple)-[~/…/data/data/com.whatsapp/databases]
└─$ ls
axolotl.db      axolotl.db-wal   companion_devices.db  hsmpacks.db               location.db      location.db-wal  media.db-shm  msgstore.db      msgstore.db-wal  payments.db-shm  statusranking.db  wa.db      wa.db-wal
axolotl.db-shm  chatsettings.db  emojidictionary.db    _jobqueue-WhatsAppJobManager  location.db-shm  media.db         media.db-wal  msgstore.db-shm  payments.db      payments.db-wal  stickers.db       wa.db-shm

┌──(water㉿kalipurple)-[~/…/data/data/com.whatsapp/databases]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/data/data/com.whatsapp/databases]
└─$ ls
axolotl.db      axolotl.db-wal   companion_devices.db  hsmpacks.db               location.db      location.db-wal  media.db-shm  msgstore.db      msgstore.db-wal  payments.db-shm  statusranking.db  wa.db      wa.db-wal
axolotl.db-shm  chatsettings.db  emojidictionary.db    _jobqueue-WhatsAppJobManager  location.db-shm  media.db         media.db-wal  msgstore.db-shm  payments.db      payments.db-wal  stickers.db       wa.db-shm

┌──(water㉿kalipurple)-[~/…/data/data/com.whatsapp/databases]
└─$ sqlitebrowser msgstore.db
```



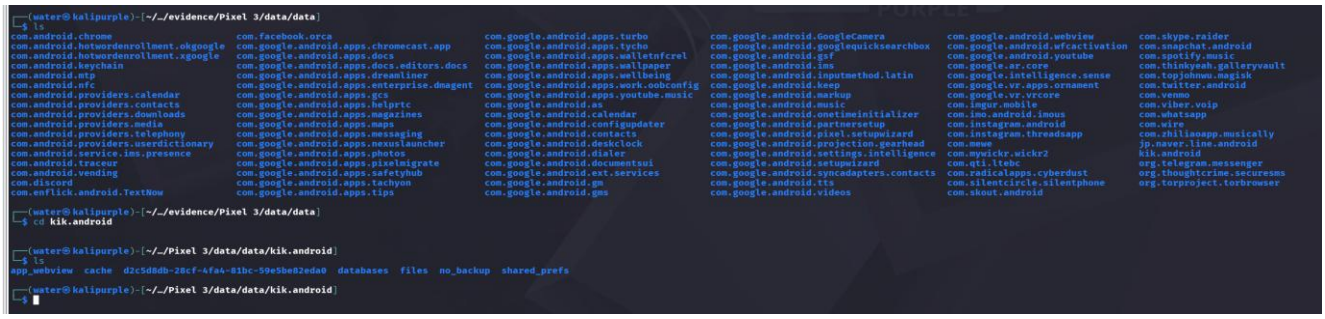

```
┌──(water㉿kalipurple)-[~/…/data/data/com.whatsapp/databases]
└─$ sqlite3 msgstore.db
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> PRAGMA table_info(jid);
```

cid	name	type	notnull	dflt_value	pk
0	_id	INTEGER	0	NULL	1
1	user	TEXT	1	NULL	0
2	server	TEXT	1	NULL	0
3	agent	INTEGER	0	NULL	0
4	device	INTEGER	0	NULL	0
5	type	INTEGER	0	NULL	0
6	raw_string	TEXT	0	NULL	0

```
sqlite>
```

Investigating Kik

Kik Database





```
┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/kik.android]
└─$ cd shared_prefs

┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$ ls
admob.xml                                          d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.google.android.gms.measurement.prefs.xml  kinecosystem_account_manager.xml
BNC_Server_Request_Queue.xml                       d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.kik.android.registerSharedPrefs.xml       kinecosystem_blockchain_source.xml
branch_referral_shared_pref.xml                    d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.joshuahickman1_ulc@talk.kik.com.xml  kinecosystem_sign_in_pref.xml
com.crashlytics.prefs.xml                          d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.xml                           KinKeyStore_kinecosystem_store.xml
'com.crashlytics.sdk.android&#58;answers&#58;settings.xml'  d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContactImageCache.pref.xml       mopubSettings.xml
com.google.android.gms.appid.xml                   d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContentImageCache.pref.xml       TwitterAdvertisingInfoPreferences.xml
com.google.android.gms.measurement.prefs.xml      'io.fabric.sdk.android&#58;fabric&#58;io.fabric.sdk.android.Onboarding.xml'         WebViewChromiumPrefs.xml
com.mopub.privacy.xml                              kik.android_preferences.xml
com.mopub.settings.identifier.xml                  KikUltraPersistence.xml

┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$ ll
total 116
-rw-rw-r-- 1 water water   257 Feb  9  2020  admob.xml
-rw-rw-r-- 1 water water   123 Feb  9  2020  BNC_Server_Request_Queue.xml
-rw-rw-r-- 1 water water  2049 Feb  9  2020  branch_referral_shared_pref.xml
-rw-rw-r-- 1 water water   251 Jan 29  2020  com.crashlytics.prefs.xml
-rw-rw-r-- 1 water water   125 Jan 29  2020 'com.crashlytics.sdk.android&#58;answers&#58;settings.xml'
-rw-rw-r-- 1 water water  2514 Feb  9  2020  com.google.android.gms.appid.xml
-rw-rw-r-- 1 water water  1001 Feb 14  2020  com.google.android.gms.measurement.prefs.xml
-rw-rw-r-- 1 water water   932 Jan 29  2020  com.mopub.privacy.xml
-rw-rw-r-- 1 water water   377 Feb  9  2020  com.mopub.settings.identifier.xml
-rw-rw-r-- 1 water water   441 Jan 29  2020  d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.google.android.gms.measurement.prefs.xml
-rw-rw-r-- 1 water water    65 Jan 29  2020  d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.com.kik.android.registerSharedPrefs.xml
-rw-rw-r-- 1 water water   160 Feb  9  2020  d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.joshuahickman1_ulc@talk.kik.com.xml
-rw-rw-r-- 1 water water 18943 Feb 14  2020  d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.KikPreferences.xml
-rw-rw-r-- 1 water water   140 Jan 29  2020  d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContactImageCache.pref.xml
-rw-rw-r-- 1 water water   140 Jan 29  2020  d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.Kik.Storage.ContentImageCache.pref.xml
-rw-rw-r-- 1 water water   168 Jan 29  2020 'io.fabric.sdk.android&#58;fabric&#58;io.fabric.sdk.android.Onboarding.xml'
-rw-rw-r-- 1 water water  2661 Jan 29  2020  kik.android_preferences.xml
-rw-rw-r-- 1 water water  2598 Feb 14  2020  KikUltraPersistence.xml
-rw-rw-r-- 1 water water   113 Jan 29  2020  kinecosystem_account_manager.xml
-rw-rw-r-- 1 water water   348 Jan 29  2020  kinecosystem_blockchain_source.xml
-rw-rw-r-- 1 water water   714 Jan 29  2020  kinecosystem_sign_in_pref.xml
-rw-rw-r-- 1 water water   359 Jan 29  2020  KinKeyStore_kinecosystem_store.xml
-rw-rw-r-- 1 water water   187 Jan 29  2020  mopubSettings.xml
-rw-rw-r-- 1 water water   213 Jan 29  2020  TwitterAdvertisingInfoPreferences.xml
-rw-rw-r-- 1 water water   127 Jan 29  2020  WebViewChromiumPrefs.xml

┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$
```

Checking Installation timestamp

```
┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$ cat com.google.android.gms.measurement.prefs.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<map>
    <long name="time_active" value="16946" />
    <long name="last_upload" value="1580322534556" />
    <long name="first_open_time" value="1580322534533" />
    <string name="gmp_app_id">1:7203525089:android:c7f7cd15833ee205</string>
    <string name="app_instance_id">3e215f4b36b0d6baf2483569077efbc3</string>
    <boolean name="app_backgrounded" value="true" />
    <long name="health_monitor:start" value="1581692483339" />
    <boolean name="use_service" value="true" />
    <boolean name="deferred_analytics_collection" value="false" />
    <long name="app_install_time" value="1580317722000" />
    <string name="previous_os_version">10</string>
    <boolean name="has_been_opened" value="true" />
    <boolean name="use_dynamite_api" value="false" />
    <boolean name="allow_remote_dynamite" value="false" />
    <long name="last_pause_time" value="1581270799473" />
    <boolean name="start_new_session" value="false" />
</map>

┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$
```

```
┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$ date -d '@1580317722000' -u
Fri Apr 10 09:00:00 AM UTC 52048

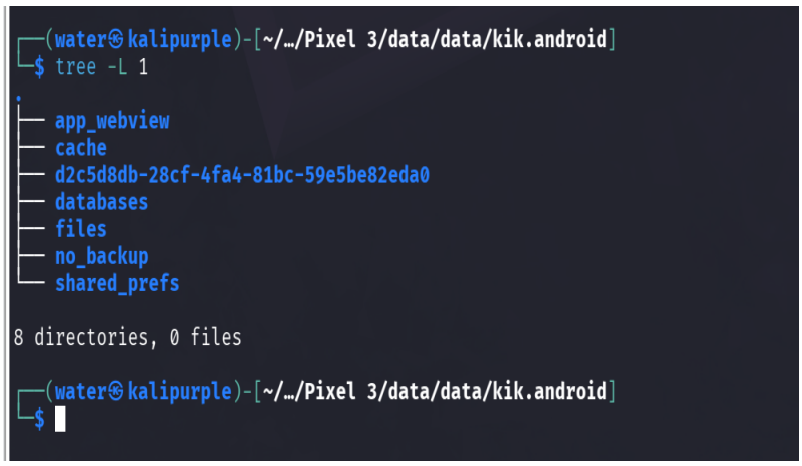
┌──(water㉿kalipurple)-[~/…/data/data/kik.android/shared_prefs]
└─$
```

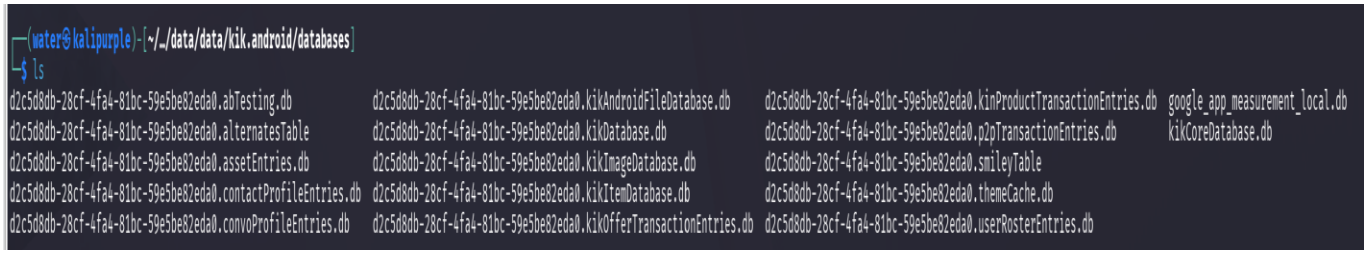
Kik Version



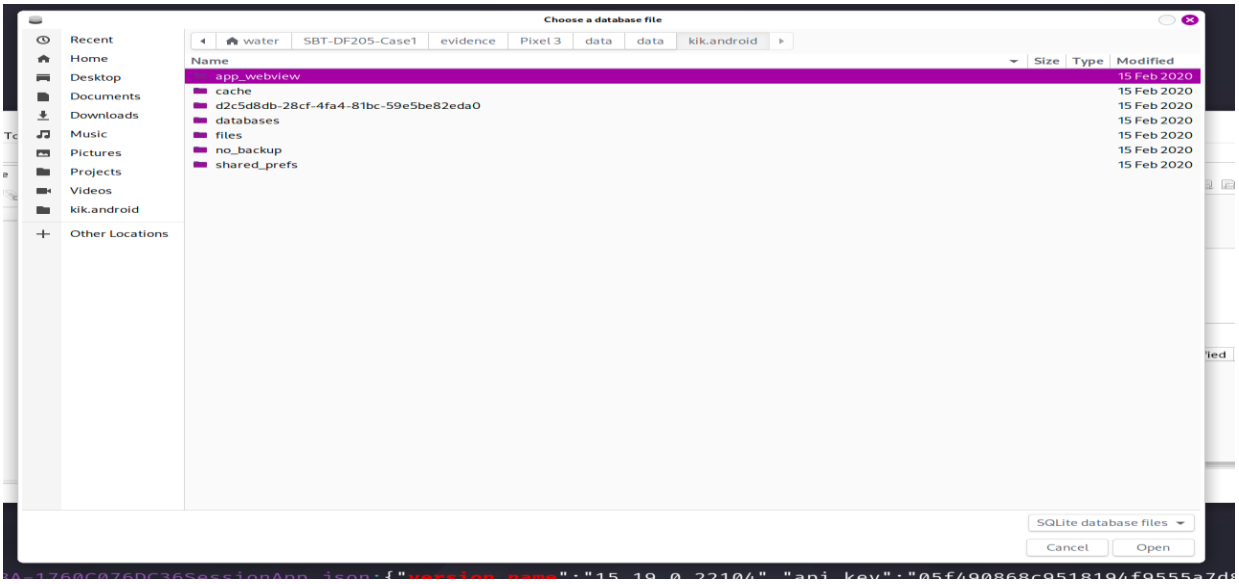


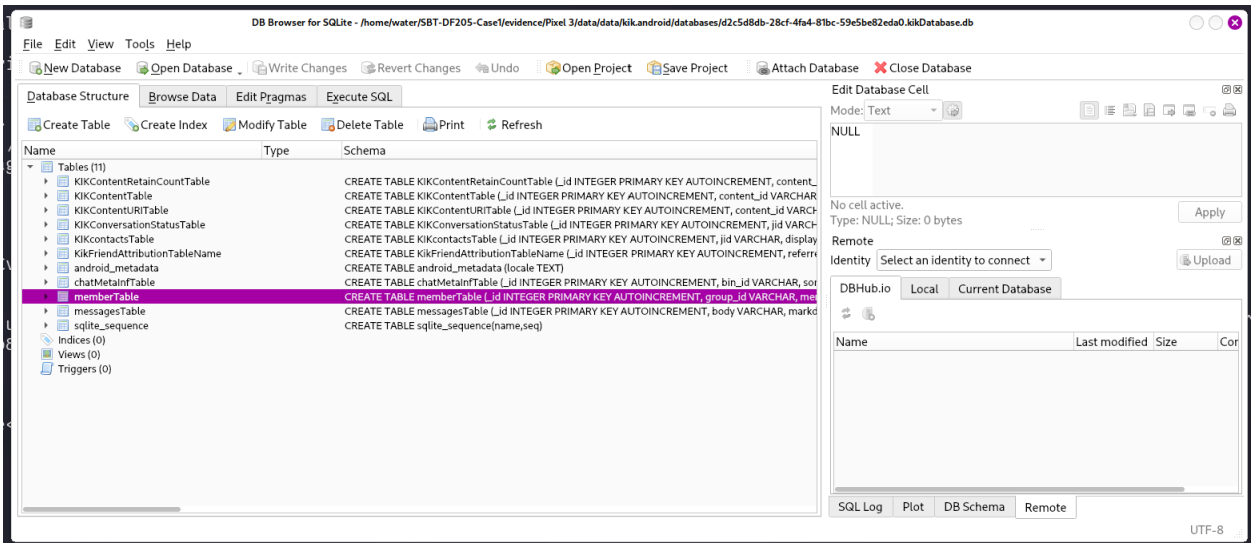
Location where conversation are stored/Database Table

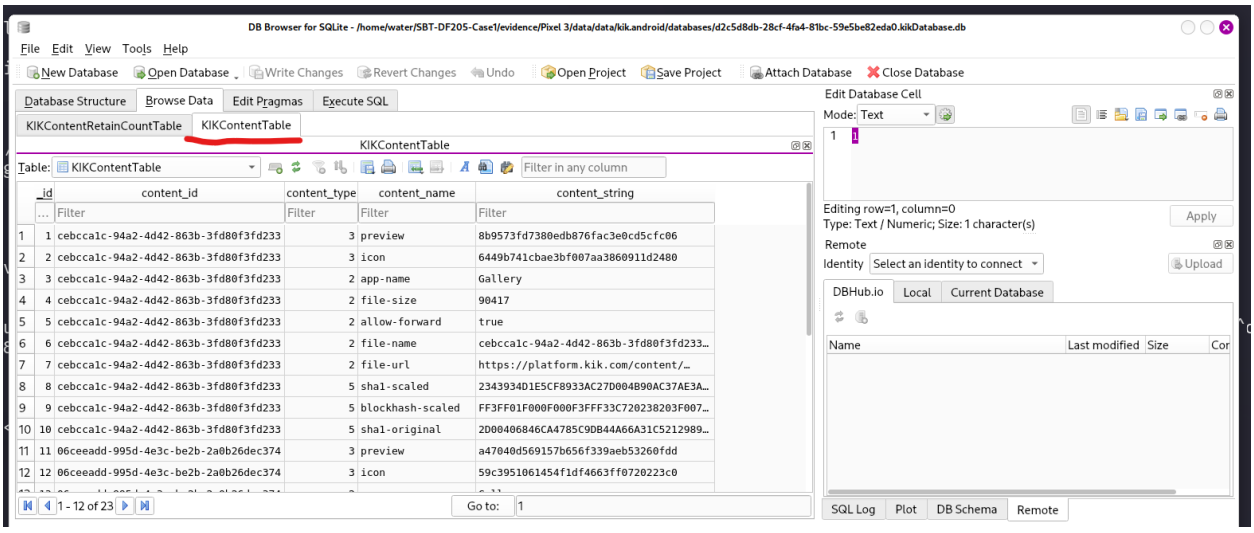







Choose a database file
Recent
Home
Desktop
Documents
Downloads
Music
Pictures
Projects
Videos
kik.android
Other Locations
water
SBT-DF205-Case1
evidence
Pixel 3
data
data
kik.android
Name
Size
Type
Modified
app_webview 15 Feb 2020
cache 15 Feb 2020
d2c5d8db-28cf-4fa4-81bc-59e5be82eda0 15 Feb 2020
databases 15 Feb 2020
files 15 Feb 2020
no_backup 15 Feb 2020
shared_prefs 15 Feb 2020
SQLite database files
Cancel
Open


DB Browser for SQLite - /home/water/SBT-DF205-Case1/evidence/Pixel 3/data/data/kik.android/databases/d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikDatabase.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
Create Table
Create Index
Modify Table
Delete Table
Print
Refresh
Name
Type
Schema
Tables (11)
KIKContentRetainCountTable
KIKContentTable
KIKContentURITable
KIKConversationStatusTable
KIKContactsTable
KikFriendAttributionTableName
android_metadata
chatMetaInfTable
memberTable
messagesTable
sqlite_sequence
Indices (0)
Views (0)
Triggers (0)
Edit Database Cell
Mode: Text
NULL
No cell active.
Type: NULL; Size: 0 bytes
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote
UTF-8


DB Browser for SQLite - /home/water/SBT-DF205-Case1/evidence/Pixel 3/data/data/kik.android/databases/d2c5d8db-28cf-4fa4-81bc-59e5be82eda0.kikDatabase.db
File Edit View Tools Help
New Database
Open Database
Write Changes
Revert Changes
Undo
Open Project
Save Project
Attach Database
Close Database
Database Structure
Browse Data
Edit Pragmas
Execute SQL
KIKContentRetainCountTable
KIKContentTable
KIKContentTable
Table: KIKContentTable
Filter in any column
_id
content_id
content_type
content_name
content_string
1 1 cebcca1c-94a2-4d42-863b-3fd80f3fd233 3 preview 8b9573fd7380edb876fac3e0cd5cfc06
2 2 cebcca1c-94a2-4d42-863b-3fd80f3fd233 3 icon 6449b741cbae3bf007aa3860911d2480
3 3 cebcca1c-94a2-4d42-863b-3fd80f3fd233 2 app-name Gallery
4 4 cebcca1c-94a2-4d42-863b-3fd80f3fd233 2 file-size 90417
5 5 cebcca1c-94a2-4d42-863b-3fd80f3fd233 2 allow-forward true
6 6 cebcca1c-94a2-4d42-863b-3fd80f3fd233 2 file-name cebcca1c-94a2-4d42-863b-3fd80f3fd233…
7 7 cebcca1c-94a2-4d42-863b-3fd80f3fd233 2 file-url https://platform.kik.com/content/…
8 8 cebcca1c-94a2-4d42-863b-3fd80f3fd233 5 sha1-scaled 2343934D1E5CF8933AC27D004B90AC37AE3A…
9 9 cebcca1c-94a2-4d42-863b-3fd80f3fd233 5 blockhash-scaled FF3FF01F000F000F3FFF33C720238203F007…
10 10 cebcca1c-94a2-4d42-863b-3fd80f3fd233 5 sha1-original 2D00406846CA4785C9DB44A66A31C5212989…
11 11 06ceeadd-995d-4e3c-be2b-2a0b26dec374 3 preview a47040d569157b656f339aeb53260fdd
12 12 06ceeadd-995d-4e3c-be2b-2a0b26dec374 3 icon 59c3951061454f1df4663ff0720223c0
1 - 12 of 23
Go to: 1
Edit Database Cell
Mode: Text
Editing row=1, column=0
Type: Text / Numeric; Size: 1 character(s)
Apply
Remote
Identity
Select an identity to connect
Upload
DBHub.io
Local
Current Database
Name
Last modified
Size
SQL Log
Plot
DB Schema
Remote

Task 7 Maps and photographs

1. Identify at least two Google Maps location-related entries where available. Record their source and explain whether they represent a search, saved place, route or another record type.

2. Examine at least three photographs or media records using Google Photos databases and local files. Record paths, hashes, capture-related times, available offsets and GPS metadata.

3. Attempt one comparison between Maps and media evidence. Explain its strength or limitations. Saved places and photo metadata alone do not prove a person visited a location.

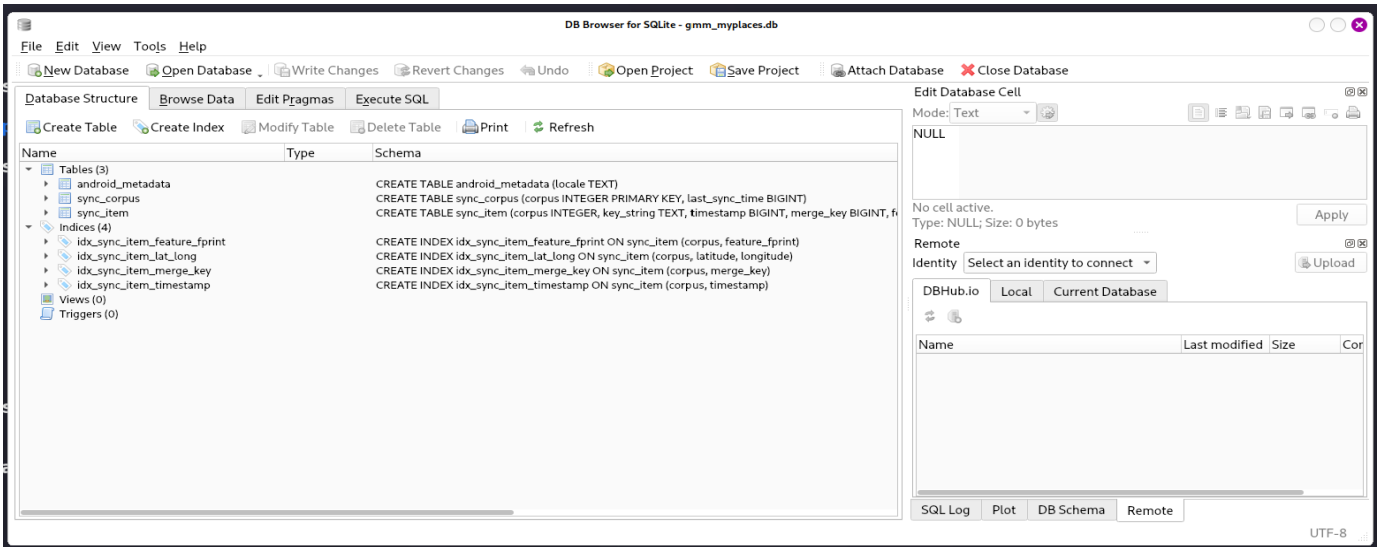


```
┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/com.google.android.apps.maps]
└─$ cd databases

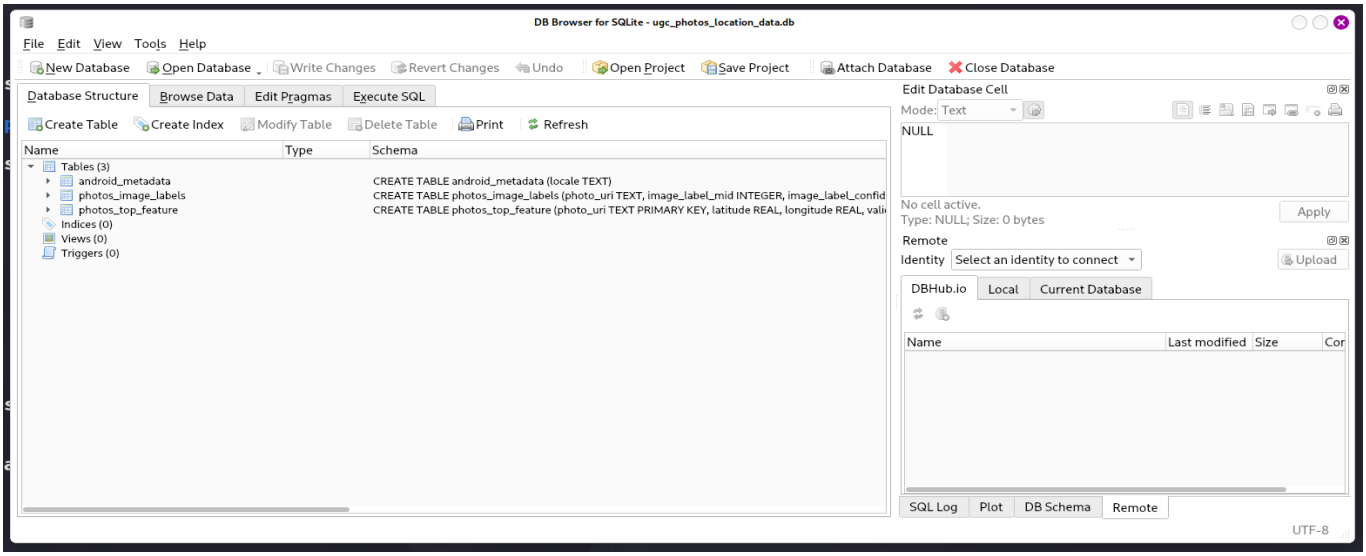
┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.maps/databases]
└─$ ll
total 248
-rw-rw-r-- 1 water water 40960 Feb  9  2020 gmm_myplaces.db
-rw-rw-r-- 1 water water 94208 Feb 14  2020 gmm_storage.db
-rw-rw-r-- 1 water water 45056 Feb  9  2020 gmm_sync.db
-rw-rw-r-- 1 water water 20480 Jan 30  2020 inbox_notifications.db
-rw-rw-r-- 1 water water 24576 Feb  9  2020 ue3.db
-rw-rw-r-- 1 water water 28672 Jan 29  2020 ugc_photos_location_data.db

┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.maps/databases]
└─$
```

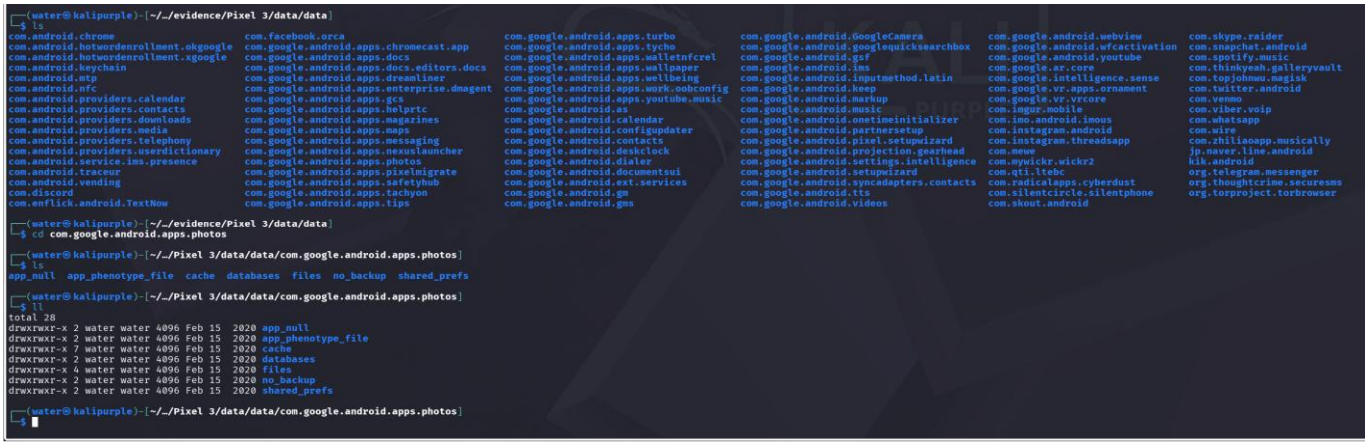
```
┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.maps/databases]
└─$ sqlitebrowser gmm_myplaces.db
```



```
┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.maps/databases]
└─$ sqlitebrowser ugc_photos_location_data.db
```



Checking Google Photo

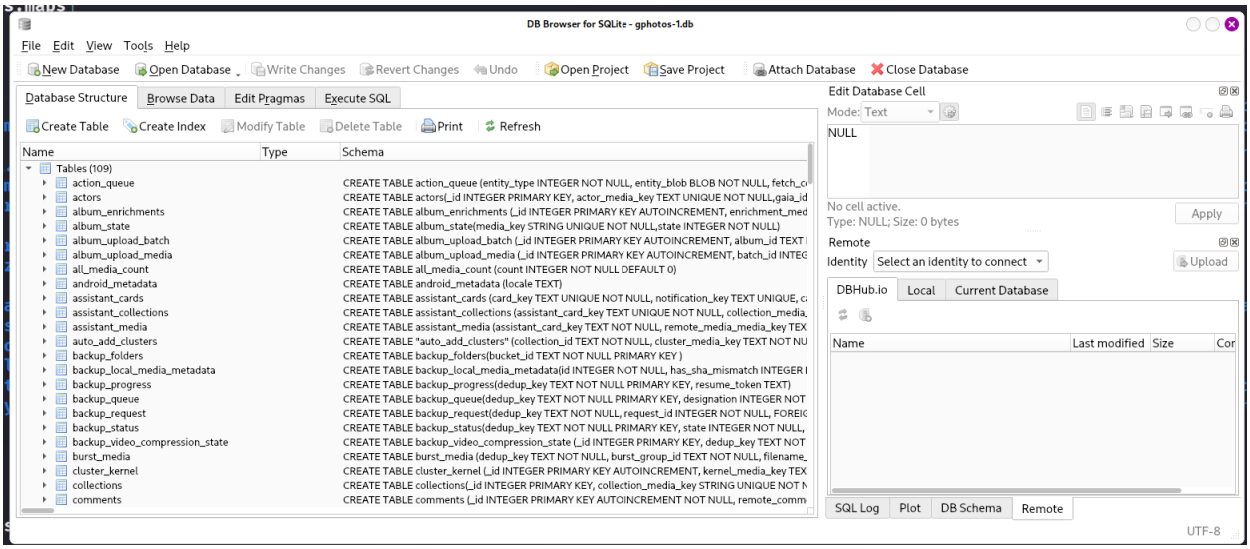



```
┌──(water㉿kalipurple)-[~/…/Pixel 3/data/data/com.google.android.apps.photos]
└─$ cd databases

┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.photos/databases]
└─$ ls
disk_cache      disk_cache-wal                  gphotos0.db      gphotos0.db-wal  gphotos-1.db-shm  local_trash.db      media_store_extras-shm  NotificationsDB0
disk_cache-shm  google_app_measurement_local.db  gphotos0.db-shm  gphotos-1.db     gphotos-1.db-wal  media_store_extras  media_store_extras-wal

┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.photos/databases]
└─$
```

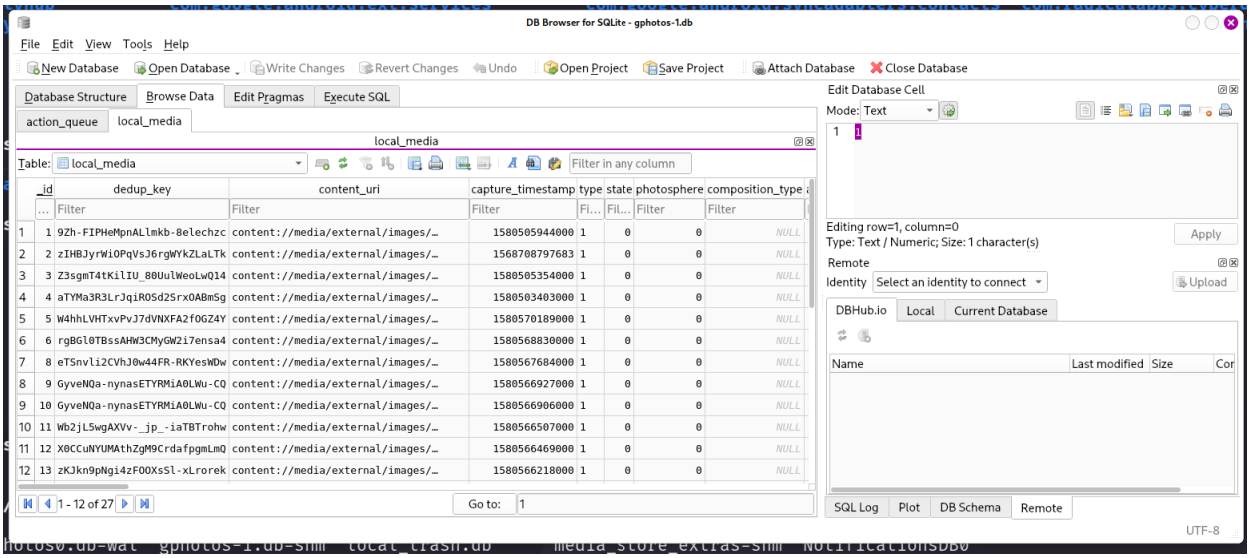
```
┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.photos/databases]
└─$ sqlitebrowser gphotos-1.db
```

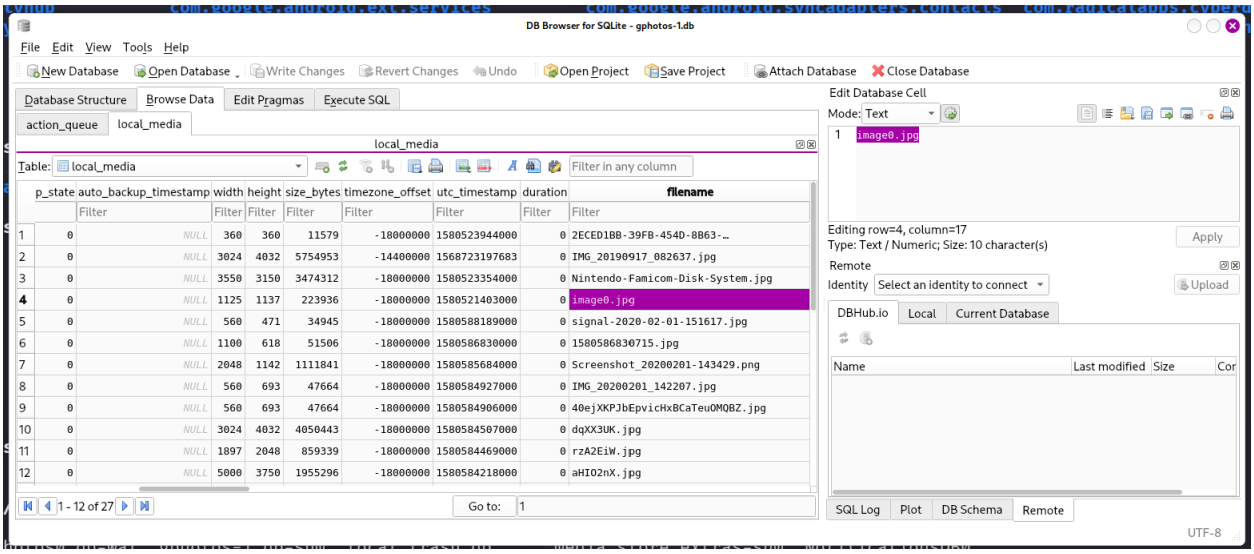



```
┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.photos/databases]
└─$ hashdeep gphotos-1.db
%%%% HASHDEEP-1.0
%%%% size,md5,sha256,filename
## Invoked from: /home/water/SBT-DF205-Case1/evidence/Pixel 3/data/data/com.google.android.apps.photos/databases
## $ hashdeep gphotos-1.db
##
1232896,9111a407d92f966f2e8604671517f900,81b58e8ea3f7224a4573e2a517fe813fb8bc329e9e388c62596412fa4a83da9e,/home/water/SBT-DF205-Case1/evidence/Pixel 3/data/data/com.google.android.apps.photos/databases/gphotos-1.db

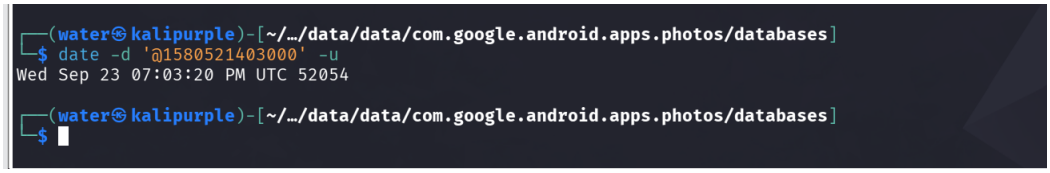
┌──(water㉿kalipurple)-[~/…/data/data/com.google.android.apps.photos/databases]
└─$
```

Samples of Pictures in Local Media





Timestamp



Task 8 Timeline and conclusions

1. Create a timeline of at least 12 distinct events from four or more artefact families. Include raw time, field name, unit or format, UTC conversion, event description and evidence ID.
2. Present at least three correlations across sources. Count duplicate representations of the same message or call as one event.
3. Conclude whether the allegation is supported, contradicted or unresolved. Explain at least two alternative interpretations and identify the additional evidence needed.

Time handling. Validate timestamp units and time zones before conversion. The creator's activity documentation uses UTC-05:00. Keep media times without a known offset explicitly unresolved.

2.7. Analysis and Findings

The investigation focused on examining the Google Pixel 3 Android 10 device for evidence of suspected unauthorised communication and information sharing. SQLite was used to examine communication databases, contacts, call logs, and application records. The examination also involved reviewing device details, user profiles, account associations, and connectivity settings. The available evidence was intended to help reconstruct relevant activities and determine whether the allegation was supported, contradicted, or unresolved.

2.8. Challenges and Solutions

One major challenge was the large size of the Google Pixel 3 evidence archive, which required sufficient storage space for extraction and analysis, so I have to install Kali purple with large disk space to do the assisgnment.Another challenge was understanding the different database structures and identifying relevant records from multiple application sources. Some timestamps also required careful interpretation to avoid incorrect time conversions.

2.9. Conclusion

The investigation provided a structured approach to examining digital evidence from a Google Pixel 3 running Android 10. The examination covered device identity, user profiles, communication records, application databases, photographs, and location-related information. Correlating evidence from different sources can help establish a clearer timeline of relevant activities. However, the presence of an account, message, photograph, or location record alone does not conclusively identify the person responsible for an activity.

2.10. Recommendations

Assignment like this improved student knowledge and understanding on Forensic case giving the student the ability to think deep and do deep research before concluding on a case

2.11. References

Class materials Guide from Founder ICDFA on Basic Networking Skills for Digital Forensics, (MobileoT Forensic Case Android 10)

https://onedrive.live.com/?id=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments%2FDocuments%2FDigital%20Forensics%2FMobileIoT%20Forensics%20Case%20Study%2FAndroid10&listurl=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments&ithint=folder&e=1YlJsb&migratedtospo=true&redeem=aHR0cHM6Ly8xZHJ2Lm1zL2YvYy9hNjc0OTA4YTVlMmUyZjJhL0lnRDVILWNwemxQQlFibjk4ekwxVjB6UkFXcEwwMXBGWGRTaVhfUkJjS2sxemJJP2U9MVlsSnNi&ga=1